

The Discrete Symmetry Ledger of the Standard Model

C, P, T, CP, Sector-Localized Violation, Invariant Diagnostics, and Normal-Form Closure

Kernel-Faithful, CPT-Synchronized, Basis-Quotiented, and Vacuum-Aware Closure with Strong- θ Control

A fixed-target classification of discrete action, representative equivalence, physical violation, anomaly, vacuum realization, and downstream standing.

Publication theorem

For every role-realization-complete closed target D_λ in the declared SYM-4 family, independently derived standing burdens and role-blind physical objects determine one lawful normal-form isomorphism class:

$$|\pi_0 \mathcal{N}_4(D_\lambda)| = 1.$$

The public sector-indexed C/P/T/CP ledger is the canonical presentation of that class. The executable certificate closes 19 targets, classifies 147 role positions against 174 presentation residues, evaluates 1311 role-residue pairs, and passes 4762 registered audit tests.

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Research article with canonical atlas, evidence archive, executable finite certificate, and Technical Audit Companion

Abstract

The expressions charge conjugation, parity, time reversal, and CP are often compressed into four labels attached to the Standard Model as a whole. That compression is not physically faithful. A named operation can denote an active map on states, a transformation of local fields, a condition on couplings, a passive convention change, a theorem consequence, a vacuum stabilizer, a parameter-limit regularity, or an invariant diagnostic. Its status can change with interaction sector, channel, field content, mass degeneracy, neutrino convention, topological sector, boundary data, operator order, and the selected vacuum. A theory-wide yes/no bit therefore erases the distinctions needed to say where and how discrete symmetry is preserved or violated.

This article develops SYM-4 of the Standard Model symmetry arc. It constructs a sector-indexed C/P/T/CP ledger and proves a fixed-target normal-form theorem for a declared family of Reference-complete targets. The logical starting point is that a determinate claim already requires fixed target identity, lawful comparison, consequence-bearing use, and irreversible fixation; those functions instantiate Admissibility, Standing, Reference, and Irreversibility. The evidentiary order nevertheless remains quantum-field-theoretic first: the operation, state or field implementation, physical quotient, interaction sector, invariant certificate, anomaly, vacuum realization, boundary/global-form dependence, and scale or operator range are fixed before any AASC role is reported.

The physical analysis separates active from passive transformations; unitary from antiunitary implementation; exact sector symmetry from parameter-limit symmetry; complex matrix representatives from physical phases; weak-basis and rephasing freedom from quotient-stable invariants; explicit from spontaneous CP violation; the QCD θ coordinate from the anomaly-invariant combination $\bar{\theta} = \theta + \arg \det M_q$ in the adopted convention; antiunitary T from naive momentum/spin reversal; direct T tests from kinematic T-odd correlations with absorptive ancestry; Dirac oscillation CP from Majorana phase loci; and the one-Higgs-doublet core from extended scalar target updates. The full Higgs Structural Endpoint Arc is imported as a load-bearing interface: the bridge/fixation carrier fixes Higgs-domain identity, while field coordinates, gauge-fixed vev phases, vacuum routes, scales, and extra scalar sectors remain separately typed.

The finite certificate uses two independently constructed upstream systems. Target primitives contain opaque physical slots and executable theorem queries; standing burdens are generated by denying one target interface and recording the resulting loss of theorem-bearing consequences. Physical candidates are generated separately from QFT seeds by a registry of 45 formula-bearing constructors executed to a fixed point of depth 5. Eligibility is operational: a residue is inserted into a denied target hole and the complete query vector is rerun. Compatibility is likewise operational: every complete eligible occupation is reassembled into a target and all local and cross-object equations are recomputed. Presentation equivalence is generated only by executable invertible maps or exact-identity certificates, with full target fingerprints defined on the resulting presentation orbits. Lawful automorphisms are computed from the full colored certificate rather than supplied as preferred permutations.

The closed family contains 19 physical targets, one separately typed integrated-ledger assembly, and 6 explicit frontier or successor records. Executed denial analysis yields 147 role positions; role-blind construction yields 213 raw artifacts and 174 presentation residues. All 1311 same-fiber burden-residue pairs and all 22 complete pointwise-eligible maps are evaluated. The full certificate contains 23 lawful automorphisms and 20 assignment orbits: one for each closed physical target and one compiler orbit for the assembly. Thus, at the declared scope,

$$\forall D_\lambda \in \Lambda_4^{\text{closed}}, \quad |\pi_0 \mathcal{N}_4(D_\lambda)| = 1.$$

The result establishes a unique lawful discrete-symmetry normal form modulo witnessed presentation. It does not predict CKM or PMNS parameters, determine $\bar{\theta}$, solve strong CP, derive baryogenesis, predict electric dipole moments, close extended-Higgs or SMEFT sectors, or claim faithful global-form and generalized-defect exhaustion. Those remain explicit successor burdens.

Read This First: Dependency and Proof Spine

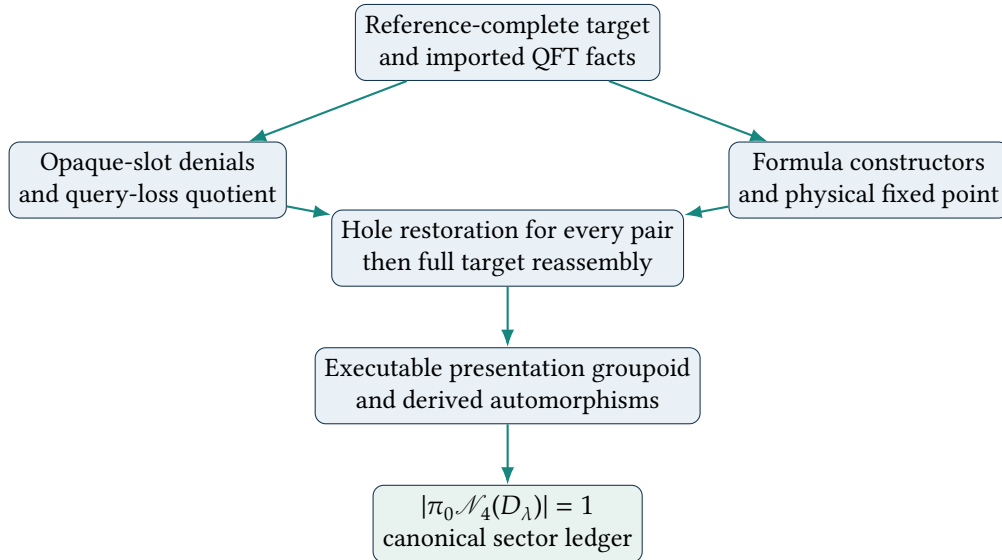
Dependency lock: kernel-first in necessity, QFT-first in certification

AASC is not appended to a standing-neutral physical ontology. The ability to keep one target, one comparison class, one claim, and one completed report fixed already instantiates the kernel. But each public physical record is certified before its AASC role is reported. This prevents a role label from manufacturing the transformation, quotient, anomaly, vacuum, or invariant facts that are later used to classify it.

The proof spine is

$$\begin{aligned}
 \text{NonDegDet}(D_\lambda^{\text{phys}}) &\implies K_{D_\lambda} = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}), \\
 K_{D_\lambda} + C_\lambda^{\text{QFT}} &\implies C_\lambda^{\text{phys}}, \\
 D_\lambda + Q_\lambda &\implies B_\lambda \implies R_\lambda = B_\lambda / \sim_{\text{gov}}, \\
 D_\lambda + C_\lambda^{\text{phys}} &\implies F_\lambda = \text{Fix}(\mathcal{K}_4) \implies X_\lambda = F_\lambda / \sim_{\text{pres}}, \\
 (R_\lambda, X_\lambda, C_\lambda) &\implies E_\lambda, J_\lambda, \Gamma_\lambda^{\text{lawful}}, \mathcal{N}_4(D_\lambda), \\
 &\implies |\pi_0 \mathcal{N}_4(D_\lambda)| = 1.
 \end{aligned}$$

Here Q_λ is the executable target-query family. The conventional physical status endpoint is completed at this QFT-certification layer, before role occupation but inside an already kernel-bearing domain. A raw burden is an observed query-loss profile produced by denying one opaque target slot. A physical artifact is a formula-bearing constructor output produced without burden input. The two systems first meet when an artifact is inserted into a denied target and the lost query vector is rerun.



Publication endpoint

For every role-realization-complete closed target in the registered SYM-4 family:

1. target identity, sector, channel, parameter region, vacuum, topology, operator category, and neutrino convention are fixed before classification;
2. target burdens are generated from executed denials and physical objects are generated by a separate formula-bearing constructor process;

3. no primitive record contains a role alias, intended occupant, compatibility verdict, or preferred assignment;
4. presentation equivalence is witnessed by executable maps with verified inverses and composition;
5. every burden-residue pair is evaluated by target-hole restoration, not by shared semantic labels;
6. every complete eligible occupation is reassembled and checked by the full target-query and cross-equation bundle;
7. the complete conventional physical status record is reportable before role occupation and remains independent of the eventual canonical assignment;
8. the induced action of the full-certificate automorphism group leaves one assignment orbit;
9. sector projections recompose to the integrated ledger, while new field content, global form, operator order, or vacuum data route to explicit successor targets.

Exact falsification routes

The result is reopened by a same-target demonstration of a missing interaction sector or channel; an omitted applicable physical constructor; an overbroad or incomplete presentation quotient; an incomplete CP-invariant certificate; a false anomaly, CPT, vacuum, Higgs, or neutrino transport; a missing denial burden; a false restoration edge; an omitted reassembly equation; an invalid map or inverse; an unaccounted lawful automorphism; a second assignment orbit; or a source proposition whose hypotheses do not match the registered target. Premise deletion, target change, or preference for another vocabulary is not by itself a same-scope counterexample.

Exact theorem boundary

The finite checker proves the normal-form result over the supplied target facts. It does not rederive the CPT theorem, the chiral anomaly, weak-basis invariant theory, nonperturbative QCD, neutrino oscillation theory, or Higgs-sector quantum field theory from empty premises. Those results remain imported with exact claim status and proposition-level source paths.

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Part I

Scope, Dependency, and Construction Discipline

Dependency Order, Necessary Kernel Activation, and Proof Class

1.1 AASC necessity, not interpretive addition

A discrete-symmetry claim is determinate only if the same target, operation, acted-upon object, comparison class, physical quotient, invariant consequence, scope and reportable endpoint persist through the classification act. If the target changes from the renormalizable Standard Model to SMEFT, if a massless-neutrino target becomes a Majorana target, if a one-doublet scalar sector becomes a two-doublet sector, or if a local Lie-algebra theory becomes a defect-completed global-form target, the claim has changed. If a basis convention changes while every physical invariant and continuation remains fixed, the claim has not changed. The ability to make those distinctions is not an optional interpretive overlay. It instantiates Reference, Admissibility, Standing and Irreversibility in the sense developed by the kernel and fixed-domain papers [Maley, 2026a,b,c].

The denial costs are direct. Without Reference, a complex matrix entry in one weak basis can be substituted for a physical CP phase in another. Without Admissibility, an axial redefinition, anomaly cancellation, vacuum update, extra scalar and change of target are not separable. Without Standing, a formal transformation law has no fixed theorem-bearing use. Without Irreversibility, later information about global form, a Majorana mass or a second Higgs doublet can be projected backward into an already completed classification and called a repair of the same act.

SYM4-T01 – Pre-role non-degenerate determinacy

For every target reported as closed, SYM-4 fixes the complete physical target fingerprint and constructs at least one standing-positive and one standing-negative completed report before the role architecture or assignment calculation is introduced. The comparison uses one fixed discriminator and changes no target-defining coordinate. The gate is therefore neither empty nor universal, and its exclusion has physical or theorem-bearing consequence.

Claim status. Local physical theorem, witnessed separately in each closed fiber.

For example, the generic three-generation quark target admits a report in which the complete CP-odd invariant certificate is nonzero and a report in which the same fixed target is falsely declared CP conserving. The invariant calculation distinguishes them. The strong-CP fiber admits the anomaly-synchronized $\bar{\theta}$ report and rejects a report treating θ alone as the physical coordinate. The one-doublet Higgs fiber admits the gauge-invariant vacuum classification and rejects a report promoting a gauge-fixed vev phase to spontaneous CP standing. These comparisons are established before any occupant is selected.

SYM4-T02 – Necessary kernel activation

For every closed SYM-4 target D_λ ,

$$\text{NonDegDet}(D_\lambda^{\text{phys}}) \implies K_{D_\lambda} = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}).$$

The local witness establishes the antecedent; the imported AASC theorem supplies the necessity implication. No sector matrix, invariant formula, finite assignment count, source-ledger entry, Higgs theorem or apex theorem is an additional activation premise.

Claim status. Imported AASC necessity theorem with local physical instantiation. The machine evidence path registers P-KERNEL - NECESSITY, sourced to Theorem 2.34 of the kernel manuscript, together with the target-specific

pre-role physical determinacy record.

Kernel activation is complete before software occupation

The machine-readable certificate does not create the kernel and cannot remove it. The phrase *pre-role endpoint* means the completed conventional physical status record formed before final AASC role occupation; it does not mean an object prior to Admissibility, Standing, Reference, or Irreversibility. A missing or stale endpoint mirror is therefore a certificate-synchronization defect, not evidence that a determinate physical target is kernel-free. Kernel necessity remains logically prior, while the local assignment theorem and the downstream apex application retain their separate target-specific proof burdens. The dependency audit deliberately breaks the downstream presentation groupoid: SYM4-T30, T34, T39 and T40 then become open, while T01 and T02 remain established. This mutation fixes the proof direction in executable form.

1.2 Kernel-first in necessity; QFT-first in record certification

The logical order and evidentiary order are distinct. The kernel is logically prior because no target-relative physical statement is available without fixed identity and admissibility. Quantum field theory is evidentially prior to final role reporting because the physical transformation law, antiunitary or unitary implementation, weak-basis quotient, anomaly, vacuum realization and process channel must be established without consulting the desired AASC role. This separation is the main anti-circularity device of the symmetry arc [Maley, 2026d,c].

The division of labor is:

1. SYM-0 supplies the kind-indexed status calculus, target update discipline, presentation quotient and role vocabulary.
2. SYM-1 supplies the gauge architecture/operation distinction and the electroweak gauge-quotient interface.
3. SYM-2 supplies the active/passive distinction, projective unitary/antiunitary implementation, and the exact CPT premise/action relation packet.
4. SYM-3 supplies the anomaly ontology, including the distinction among gauge inconsistency, anomalous global response, counterterm transport, global obstruction and matching.
5. The Standard Model, mixing, neutrino and Higgs support corpora supply the target-specific carrier, quotient and residual-status propositions.
6. The finite certificate verifies role occupation and normal-form uniqueness over those imported physical facts.

1.3 Five sequential proof layers

The manuscript preserves five proof layers.

Layer I: necessary kernel and fixed-domain layer. Non-degenerate determinacy instantiates the kernel and forbids hidden same-act repair. This layer is not conditional on the later role assignment or on a software endpoint field.

Layer II: reference-complete physical layer. Each target fixes field content, interaction sector, background, boundary, phase, global form, operator category, neutrino convention, parameter conditions and quantum regime. Imported theorems are used only where their hypotheses match.

Layer III: independent construction layer. Target denial generates occupant-free standing burdens. A separate physical constructor grammar generates action, structure, diagnostic and transport objects without role input.

Layer IV: local normal-form layer. Presentation witnesses, all-pairs eligibility, relational compatibility and lawful automorphisms determine the assignment groupoid.

Layer V: downstream apex layer. Only after the local singleton theorem may the no-second-classifier consequence be imported. The apex result cannot choose an occupant, repair a missing physical relation or convert an incomplete target into a closed one [Maley, 2026e,f].

SYM4-T03 – Reference-complete indexed-target fixation

Every SYM-4 claim is attached to exactly one fixed target, a declared transport between fixed targets, an explicit target update, an incomplete certificate or an open frontier. A target is eligible for closure only when every coordinate used by its physical propositions, presentation witnesses, compatibility relations, continuations and failure routes is fixed and mutually consistent.

SYM4-T04 – Imported-interface identity and no status laundering

An imported proposition remains imported even when it is repeated across several corpus papers. Repetition does not convert a continuum-QFT theorem into a local proof. Conversely, a theorem-derived physical action is not demoted to presentation merely because its explanatory role is downstream. Claim status, evidence strength and AASC role are independent ledgers.

1.4 Proof class and exact burden

This article is a kernel-forced, fixed-target, proposition-certified, executable finite normal-form theorem with imported physical interfaces. It is not a derivation of the Standard Model from empty premises, not a numerical fit, not a replacement for direct CP or T experiments, and not a calculation of nonperturbative QCD. A criticism has theorem standing if it preserves the target while breaking kernel necessity, invalidating an imported physical proposition or its specialization, exposing a missing sector, burden, object, constructor, presentation witness, eligibility edge, compatibility relation or automorphism, or producing a second faithful same-target assignment orbit.

Exact nonclaim

No foundational conclusion is inferred merely from the fact that the executable audit returns PASS. PASS certifies the declared finite theorem object. It does not generate the imported physical truths on which that object is conditioned.

Integrated Sector-Indexed Target and Target-Update Discipline

2.1 The separately typed integrated ledger assembly

The integrated object is A-SYM4- INTEGRATED- LEDGER. It is a compiler and synchronization assembly over the Reference-complete physical fibers, not itself an additional closed physical theory. This distinction prevents unresolved alternatives or mutually exclusive parameter regimes from being smuggled into one closure-friendly target. The assembly fixes simultaneously:

- four-dimensional oriented and time-oriented Minkowski spacetime, the selected positive-metric state interface, the connected Poincare action and the canonical SYM-2 CPT packet;
- the anomaly-free sharp-local Standard Model gauge architecture and its physical quotient, with faithful global-form and defect completion explicitly deferred;
- the observed chiral matter representation pattern and three fermion generations as fixed-boundary envelope data;
- one Higgs doublet, the selected electroweak vacuum and the Higgs bridge/fixation carrier interface;
- renormalizable gauge, Yukawa and scalar interactions, together with exact quark, charged-lepton and target-specific neutrino ledgers;
- interaction-sector, process-channel, perturbative/nonperturbative, boundary, topology, operator-category and scale fields;
- the weak-basis, rephasing, permutation and degeneracy quotients required before a phase or mixing statement can become physical;
- target-indexed anomaly, strong- θ , CPT, vacuum and proposition-level evidence interfaces.

The physical theorem quantifier ranges over the 19 closed fibers listed below. The assembly has its own separately reported compiler normal form, obtained only after those fibers and their transport rules have been fixed. This two-level architecture is stronger than treating the assembly as a closed physical target: it preserves mutually exclusive target coordinates while still proving that their public ledger recomposes coherently.

The assembly is deliberately parameterized. It does not assume that every numerical phase or mass has a unique value. It fixes the structural slots and the comparison regime in which CP status can be compiled. The residual-parameter theorem classifies three generations and non-skin numerical values as fixed-boundary data inside the declared representation regime; it explicitly does not derive the number three or the CKM and PMNS coordinates [Maley, 2026g].

Generation-count lock

SYM-4 uses $N_g = 3$ as an immutable coordinate of the observed Standard Model envelope. Changing N_g changes the dimensions of Yukawa matrices, mixing quotients, flavor stabilizers and invariant sets. It is therefore a target update, not a same-scope parameter variation. No local SYM-4 theorem is advertised as a derivation of three generations.

2.2 Closed targets, controlled variants and frontiers

The closure theorem is fiberwise. The integrated parent is supplemented by projections and controlled variants that isolate physically distinct quotient or parameter regimes. These are not independent universes invented to force a result. Each is a fixed restriction or successor target with a declared map to the parent.

Table 2.1: Reference-complete closed physical targets.

Target	Name	Sector	Fixed scope	Roles	Residues
F-QED-VECTORLIKE-REALMASS	Vectorlike QED with real mass and no external backgrounds	QED	Reference-complete finite benchmark target	6	6
F-WEAK-CHIRAL-1 GEN	One-generation charged-current weak target with real coupling	weak charged current	Reference-complete finite benchmark target	6	6
F-WEAK-NEUTRAL-CURRENT-TREE-3 GEN	Three-generation tree-level Standard Model neutral-current target	weak neutral current	Reference-complete finite benchmark target	8	8
F-CKM-GENERIC-3GEN	Generic nondegenerate three-generation quark Yukawa target	quark flavor	Reference-complete finite benchmark target	8	11
F-CKM-CPZERO-3GEN	Three-generation quark target with vanishing physical CP phase	quark flavor	Reference-complete finite benchmark target	9	10
F-CKM-2GEN	Two-generation quark target	quark flavor	Reference-complete finite benchmark target	8	9
F-CKM-DEGENERATE-UC	Three-generation quark target with exact up-charm degeneracy	quark flavor	Reference-complete finite benchmark target	8	9
F-DEGENERATE-2 D-AUTOMORPHISM	Controlled two-axis degenerate flavor automorphism target	degenerate flavor control	Reference-complete finite automorphism stress target	7	7
F-STRONGCP-GENERIC	Generic strong-CP target away from fixed points	strong CP	Reference-complete finite benchmark target	8	10
F-STRONGCP-THETA0	Strong-CP target at bar-theta zero	strong CP	Reference-complete finite benchmark target	8	10
F-STRONGCP-THETAPI-INVARIANT	Strong-CP special point bar-theta pi with invariant vacuum branch	strong CP	Reference-complete finite benchmark target	8	10
F-STRONGCP-THETAPI-BROKEN	Strong-CP special point bar-theta pi with spontaneously broken vacuum branch	strong CP	Reference-complete finite benchmark target	8	10
F-STRONGCP-MASSLESS-U	Strong-CP target with one exactly massless quark	strong CP	Reference-complete finite benchmark target	8	10
F-HIGGS-ONE-DOUBLET	Minimal one-Higgs-doublet scalar and vacuum CP target	Higgs/vacuum	Reference-complete finite benchmark target	7	8
F-NU-MASSLESS	Minimal massless-neutrino target	neutrino/lepton	Reference-complete finite benchmark target	8	10
F-NU-DIRAC	Three-generation Dirac-neutrino target	neutrino/lepton	Reference-complete finite benchmark target	8	10
F-NU-MAJORANA-WEINBERG	Three-generation Weinberg-operator Majorana-neutrino target	neutrino/lepton	Reference-complete finite benchmark target	8	11
F-BVV-TRIPLE-FAKE	B to VV transversity target with absorptive fake T-odd correlation	decay/T-odd	Reference-complete finite benchmark target	6	9
F-BVV-TRIPLE-TRUE	B to VV transversity target with weak-phase CP/T violation	decay/T-odd	Reference-complete finite benchmark target	6	9

The controlled fibers are chosen to exercise the distinctions that most often fail in informal prose: generic versus vanishing CP invariants, two generations versus three, exact degeneracy, generic versus massless-quark strong CP, a conditional CPT relation interface versus concrete CP-violating fibers, minimal versus Dirac versus Majorana neutrino content, one-doublet versus extended scalar sectors, direct T tests versus T-odd correlations, and exact CPT premises versus target updates.

2.3 The Higgs Structural Endpoint Arc as a target interface

The attached Higgs Structural Endpoint Arc materially strengthens the vacuum and scalar-sector boundary of SYM-4. Its first paper identifies the Higgs-domain standing-bearing carrier

$$C_H = H_{\text{br/fix}},$$

as the bridge/fixation carrier required by an order-to-deformation and mass-response target [Maley, 2026h]. The second paper fixes Higgs structural identity as the unique admissible occupation of that carrier, not as scalarity, a vev, a potential minimum, a pole mass or a running coordinate [Maley, 2026i]. The third removes primitive Higgs-scale standing independent of C_H [Maley, 2026j]. The fourth classifies vacuum location, lower branches, metastability, lifetime and thermal routes as carrier-indexed route statuses and proves that route availability does not imply primitive standing [Maley, 2026k]. The fifth classifies extra scalar sectors by carrier relation and routes genuine two-doublet, portal, composite, supersymmetric and related structures to enrichment or target update [Maley, 2026l]. The sixth closes the Higgs-domain attribute architecture [Maley, 2026m].

SYM-4 imports this package with a strict division of labor. Ordinary QFT establishes that the renormalizable one-doublet scalar potential and its gauge-invariant vacuum do not contain an independent spontaneous-CP phase. The Higgs Arc determines the standing status of the field, vev, vacuum route and extension claims. It does not replace the QFT calculation, and the QFT calculation does not independently supply AASC role standing.

Higgs target transport rule

A scalar-sector CP claim belongs to the core SYM-4 target only if it preserves the one-doublet field content, the Higgs carrier C_H , the selected vacuum phase and the inherited gauge quotient. A second doublet, singlet, composite sector, portal, supersymmetric scalar package or new vacuum order parameter is an explicit successor target unless a carrier-preserving projection back to the core is supplied.

2.4 Target fingerprints

Each target has an immutable fingerprint containing at least:

$$\text{FP}(D_\lambda) = (\text{fields, interactions, background, boundary, topology, } \Gamma, \\ \text{operator category, phase, vacuum, neutrino convention,} \\ \text{perturbative range, parameter limits, source and rule versions}).$$

The implementation distinguishes a raw representative digest from the physical-point fingerprint. The representative digest may change under an admitted weak-basis transformation, rephasing, degenerate-eigenspace rotation, gauge phase, ray phase, or axial redefinition. The physical fingerprint is computed from a canonical invariant packet for the complete witnessed presentation orbit and must remain unchanged under every lawful arrow. In an exactly degenerate quark eigenspace, ordered rows are not target data; the fingerprint uses the corresponding mass-eigenspace projectors and their rephasing invariants. A proposed comparison is same-scope only if the structural coordinates and physical-point fingerprint agree or are related by an explicitly registered target transport. A genuine change of physical point is a new theorem object.

Integrated-target fixation

Every submitted SYM-4 claim has exactly one of five dispositions: a target-locus facet of a closed target; a witnessed presentation of such a facet; a derived diagnostic; an explicit target update or frontier; or an incomplete/rejected claim. No unregistered same-act repair is permitted.

2.5 Explicit exclusions from core closure

The following remain outside the core quantifier: faithful Standard Model global-form selection with genuine line and monopole sectors; special- θ vacuum branching whose answer depends on that completion; finite-group and non-invertible anomaly inventories; multi-Higgs spontaneous CP; SMEFT CP-odd bases without a fixed dimension and matching scale; axion, Peccei–Quinn and other strong-CP mechanisms; baryogenesis; dynamical gravity; and quantum-gravity discrete-symmetry conjectures. These exclusions are visible target edges rather than silent omissions.

Exact falsification route

A same-target counterexample must preserve the full fingerprint. A result obtained by adding a right-handed neutrino, a second Higgs doublet, a higher-dimension operator, a monopole sector or a new boundary condition may be important physics, but it is not a contradiction of the source target unless a faithful same-target transport is supplied.

Atomic Discrete-Symmetry Objects and the Extended SYM-0 Schema

3.1 Why familiar names are not atomic records

A label such as “CP symmetry” can hide several theorem objects: the CP operation on fields, its antiunitary or unitary state-space implementation, a generalized CP map combined with a flavor rotation, the invariance of a Lagrangian term, the invariance of the selected vacuum, a weak-basis invariant, a process asymmetry, a strong- θ coordinate, or a discrete anomaly. The parent label is useful for navigation but cannot be the unit of classification.

SYM-4 therefore decomposes a familiar claim by target, sector, channel, candidate kind and locus. The top-level kind remains the SYM-0 sum type

$$\text{CandidateKind} = \text{Action} \sqcup \text{Structure} \sqcup \text{Diagnostic} \sqcup \text{Transport}.$$

An operation on states is an Action. A target-defining CPT premise packet or Higgs carrier is Structure. A Jarlskog invariant, T-odd correlation or anomaly coefficient is Diagnostic. A weak-basis change, axial redefinition or transport between neutrino targets is Transport. No kind is forced into the vocabulary of another.

3.2 The atomic record

For a fixed target D , an atomic discrete-symmetry record is

$$\mathfrak{d}_D = (D, s, \delta, k, X, \rho, q, \mathcal{J}, \mathcal{A}, V, B, R, \Phi, \Theta, N, \text{CPT}, \sigma, E),$$

where:

D – target	Complete target fingerprint and version.
s – sector/channel	Interaction sector, process channel and perturbative order.
δ – operation	C, P, T, CP, CPT or an explicitly declared composition/generalized map.
k – kind	Action, Structure, Diagnostic or Transport.
X – acted-upon object	States, rays, local fields, operators, couplings, backgrounds, vacuum data, extended operators or theory-space coordinates.
ρ – law	Explicit transformation, including unitary/antiunitary and linear/antilinear status.
q – quotient	Gauge, basis, rephasing, ray, current-improvement or other presentation quotient.
$\mathcal{J}$ – invariant certificate	Complete target-relative invariant set, a certified sufficient witness, or an explicit incompleteness marker.
$\mathcal{A}$ – anomaly	Target-indexed SYM-3 import: none, gauge inconsistency, anomalous current, 't Hooft obstruction, global/discrete anomaly or frontier.
V – vacuum	Invariant, spontaneously realized, explicitly deformed, gauge-fixed representative, route status or not applicable.

B – boundary/global data

Boundary, topology, faithful global form, line/defect and operator-category dependence.

 R – range

Parameter region, operator order, scale and perturbative/nonperturbative range.

 Φ – flavor quotient

Weak-basis, rephasing, permutation and degeneracy-stabilizer data.

 Θ – strong phase θ , mass determinant phase, anomaly transport, periodicity and vacuum branch. **N – neutrino target**

Minimal massless, Dirac, Majorana or explicit successor target.

CPT – dependency

Exact imported premise, theorem and implementation path used by any CP–T inference.

 σ – AASC role

Anchor, carrier constraint, bridge/fixation, physical action, witness, skin, effective route, obstruction or target update.

 E – evidence

Source propositions, claim status, rivals, nonclaims, version and integrity digest.

3.3 Kind-indexed physical disposition

The quotient question is meaningful only for Actions. For an action g on X_D with physical quotient q_D ,

$$q_D(gx) = q_D(x) \quad \forall x$$

defines a quotient-trivial action; an explicit witness x with $q_D(gx) \neq q_D(x)$ establishes quotient-nontrivial action. Apparent mixed cases are split by locus. A boundary action and its proper-bulk restriction are separate records. A T-odd observable is not classified by this equation because it is a Diagnostic, not an operation.

The record also distinguishes:

- exact action versus invariant fixed points;
- classical term invariance versus quantum measure or determinant status;
- action symmetry versus vacuum symmetry;
- exact generic target versus exact parameter limit;
- state action versus field-component covariance;
- local current relation versus global charge and process asymmetry.

SYM4-T05 – Discrete-object and locus noncollapse

Discrete action, implementation, representative convention, invariant diagnostic, anomaly, vacuum realization and target update are pairwise non-identical atomic kinds. No one of them is inferred from the word C, P, T or CP alone. Two records can be merged only by an invertible target-relative witness preserving the full physical and failure tuple.

3.4 Causal ancestry

Every non-exact result is typed by cause. SYM-4 distinguishes:

1. chiral representation asymmetry;
2. a nonzero quotient-stable Yukawa or mixing invariant;

3. nonzero $\overline{\theta}$;
4. explicit scalar-potential deformation in a successor target;
5. spontaneous vacuum selection;
6. anomalous current or discrete/global anomaly;
7. absorptive or final-state phase;
8. boundary or external-background asymmetry;
9. higher-dimension operator;
10. target update.

A T-odd correlation generated by an absorptive phase and a direct mismatch between a transition and its time reverse are not assigned the same ancestry. A quark mass term and the axial anomaly are not two names for the same breaking source. A gauge-fixed Higgs phase and a gauge-invariant vacuum order parameter are not interchangeable.

3.5 Completion and fail-closed disposition

Schema completeness, physical resolution, role closure and source integrity are independent. A row can be structurally well-typed but physically open, as in the special- θ or discrete-anomaly frontier. A row with a missing invariant certificate does not default to CP violation or CP conservation. Its disposition is incomplete.

The admissible report-level outcomes are:

NormalForm, ConditionalRule, IncompleteCertificate, IncompleteTarget,
OpenFrontier, TargetUpdate, Rejected, Superseded.

SYM4-T06 – Sector projection and verified recomposition

Every atomic row is a projection of a fixed parent target or an explicit successor. Projection is lawful only when the parent fingerprint and all consumed interfaces are retained. Recomposition succeeds only when action, quotient, invariant, CPT, anomaly, vacuum and source fields agree on shared loci. A collection of individually correct sector statements constitutes an integrated Standard Model ledger only when this synchronization passes.

Exact nonclaim

The record schema does not imply that every coordinate is primitive or even physically applicable. It is a completeness discipline. “Not applicable” is a typed value; silence is not.

Independent Burden and Physical-Object Construction

4.1 Why file separation is insufficient

A closure calculation is circular when its purportedly independent inputs are merely stored in different files while carrying matching conclusion-bearing labels. The present construction removes that semantic bridge. The burden process sees only opaque target-slot identifiers and executable theorem queries. The physical process sees only QFT seeds and formula-bearing constructors. Neither process reads the other's ontology, capability vocabulary, or intended occupation.

4.2 Burden process: denials and query losses

For a fixed target D_λ , let

$$Q_\lambda(D_\lambda) = \{Q_1(D_\lambda), \dots, Q_m(D_\lambda)\}$$

be the complete executable query vector. Each query returns PASS, FAIL, UNRESOLVED, or NOT_APPLICABLE, together with a value and witness. Examples ask whether the target is fixed, whether a discrete action is defined, whether it descends to physical rays, whether the complete invariant packet agrees with the selected mass data, whether $\bar{\theta}$ is preserved by an axial map, or whether a vacuum packet is invariant.

The target packet contains opaque slots $s_1, \dots, s_n$. For each slot, the burden engine constructs the denied target $D_\lambda^{-s_i}$ and recomputes the full vector. The raw burden is

$$b_i = (L_i, \Delta_i, C_i, F_i, U_i),$$

where L_i is the set of lost queries, Δ_i records changed values, C_i the continuation effect, F_i the failure ancestry, and U_i the target-update disposition. No human role name or expected object occurs in b_i .

Raw burdens are governance-equivalent exactly when these executed profiles agree:

$$b_i \sim_{\text{gov}} b_j.$$

The role architecture is the quotient $R_\lambda = B_\lambda / \sim_{\text{gov}}$. Reader-facing names are assigned only after that quotient and are not used by the proof engine.

4.3 Physical process: seeds and formula constructors

The physical pipeline begins from seeds natural in QFT:

- discrete-operation laws on fields and states;
- projective ray and CPT interfaces;
- quark Yukawa matrices and weak-basis actions;
- QCD θ , quark masses, and axial rotations;
- the one-doublet Higgs carrier, potential, and vacuum orbit;
- massless, Dirac, and Majorana neutrino data;
- concrete $B \rightarrow V_1 V_2$ transversity amplitudes and their CP-conjugates;
- source propositions and target coordinates.

A registry of 45 formula-bearing constructors diagonalizes mass matrices, forms mixing mismatches, computes Jarlskog-type and Majorana invariants, constructs rephasing orbits, transports θ and mass phases, forms vacuum stabilizers, computes true and fake triple products, and assembles the integrated ledger. The grammar is iterated until no new typed artifact is produced:

$$F_{r+1} = F_r \cup \{\kappa(x_1, \dots, x_k) : \kappa \in \mathcal{K}_4, \text{Applicable}_\kappa\}, \quad F_{r+1} = F_r.$$

The executed fixed point contains 213 raw artifacts and reaches maximum derivation depth 5.

4.4 Semantic firewall

The primitive schemas prohibit the following fields on both sides:

`required_capabilities, allowed_candidate_kinds, eligible_role, expected_occupant,`
`canonical_assignment, physical_relations, compatibility_pass.`

The audit also blinds and permutes display aliases, removes the candidate environment while burdens are generated, and removes the burden environment while candidates are generated. Hashes of the unaffected output must remain identical.

Independent-construction theorem

For every closed target, burden formation and physical-object formation are generated from disjoint conclusion-bearing inputs. The burden side is determined by executed target-query losses; the physical side is determined by QFT seeds and formula constructors. Their first theorem-bearing contact is operational target-hole restoration.

Exact falsification route

A role alias inside a physical seed, a target slot named for its expected occupant, a primitive candidate-relation edge, or an assignment-specific PASS field invalidates the independence theorem even if the final orbit count remains one.

Conventions, Imported Interfaces, and Source Authority

5.1 Discrete-operation conventions

The manuscript fixes conventions once and records convention changes as witnessed presentations. Charge conjugation acts by the declared conjugation map on fields and states. Parity reverses spatial orientation while preserving the selected time orientation. Time reversal is an antiunitary operation on physical rays; it is not defined by changing the sign of displayed momenta and spins alone. CP and CPT are compositions whose phase conventions can change without changing the physical operation when an exact witness preserves the full tuple.

For a local field $\Phi_a(x)$, the schematic transformations are written

$$\begin{aligned} C \Phi_a(x) C^{-1} &= (\mathcal{C}\Phi)_a(x), \\ P \Phi_a(t, \mathbf{x}) P^{-1} &= (\mathcal{P}\Phi)_a(t, -\mathbf{x}), \\ T \Phi_a(t, \mathbf{x}) T^{-1} &= (\mathcal{T}\Phi)_a(-t, \mathbf{x}), \end{aligned}$$

with T antiunitary and with field-dependent matrices, intrinsic phases and index operations fixed by the target. These formulas are coordinates, not universal proofs of exactness.

5.2 Yukawa and strong-phase convention

The quark Yukawa sector is

$$\mathcal{L}_Y = -\bar{Q}_L Y_d H d_R - \bar{Q}_L Y_u \tilde{H} u_R + \text{h.c.}$$

and weak-basis changes act simultaneously on fields and matrices. Physical conclusions are stated only after quotienting that action.

For strong CP the manuscript adopts

$$\bar{\theta} = \theta + \arg \det M_q.$$

Under a common axial rotation of N_f quarks in the convention used here,

$$\theta \mapsto \theta + 2N_f \alpha, \quad \arg \det M_q \mapsto \arg \det M_q - 2N_f \alpha,$$

so $\bar{\theta}$ is invariant. A different sign convention is admissible only if the action, anomaly and mass-phase definitions are changed coherently [Adler, 1969, Bell and Jackiw, 1969, Fujikawa, 1979, Maley, 2026n,o].

5.3 Neutrino convention

Three neutrino targets are kept distinct:

1. the minimal renormalizable Standard Model with no neutrino mass term;
2. a Dirac target with right-handed neutrinos and the corresponding rephasing freedoms;
3. a Majorana target with restricted rephasings and lepton-number-violating operator loci.

The PMNS matrix is never imported without its target. Oscillation probabilities use rephasing-invariant quartets; the Majorana phase factors cancel from ordinary oscillations but can enter lepton-number-violating amplitudes [Maki et al., 1962, Schechter and Valle, 1980, Bilenky et al., 1980, Maley, 2026p,q].

5.4 Claim-status and evidence ledgers

Three independent fields are maintained:

Claim status Imported QFT, imported AASC, locally proved, executable finite theorem, grammar-relative coverage, conditional frontier, open, rejected or superseded.

Evidence strength Primary theorem, standard derivation, exact corpus proposition, worked model, controlled example or heuristic.

Confidence Project-management estimate of survival under external review; it is not theorem status.

The source policy is proposition-level. A bibliography entry alone is not an evidence certificate. Each load-bearing import records the exact target hypotheses, proposition content, local specialization, limitation, source identity and integrity status. The local archive contains 48 supplied PDFs totaling 1842 pages, each fixed by SHA-256. External QFT sources remain bibliographically controlled rather than falsely reported as locally rederived.

5.5 Primary physics authorities

The quark mixing and CP analysis imports the Cabibbo and Kobayashi–Maskawa structures and the Jarlskog weak-basis invariant criterion [Cabibbo, 1963, Kobayashi and Maskawa, 1973, Jarlskog, 1985]. Parity and CP violation are anchored by the classic weak-interaction and neutral-kaon results [Lee and Yang, 1956, Wu et al., 1957, Christenson et al., 1964]. Direct T tests are kept distinct from CPT inference and T-odd correlations [CPLEAR Collaboration, 1998, BaBar Collaboration, 2012]. Gauge/vacuum claims use Elitzur’s theorem at the appropriate locus [Elitzur, 1975]. The CPT interface is imported from SYM-2 and its exact premise ledger rather than restated as folklore [Maley, 2026r, Greaves and Thomas, 2014].

Source-matching rule

An imported result is used only at the strength of its hypotheses. A statement about generic nondegenerate three-generation quarks cannot be transported to an exact degeneracy without a new invariant certificate. A result about Dirac neutrinos cannot be transported to Majorana rephasing. A one-doublet vacuum theorem cannot be applied to a two-doublet extension. A local anomaly cancellation does not certify global-form completion.

Exact nonclaim

The use of a corpus theorem as an AASC dependency does not replace the need for a mainstream QFT certificate. Conversely, a mainstream QFT theorem does not by itself settle the AASC role occupied by the certified object.

Part II

Quantum-Field-Theoretic Discrete-Symmetry Architecture

C, P, T, CP, and CPT Implementations

6.1 Operations before invariance claims

A discrete-symmetry statement begins by specifying an operation. It does not begin with the word “violation.” The acting object must be defined on the admitted state and field spaces, and the transformation must preserve the target’s algebraic relations sufficiently to make the invariance question meaningful.

For a multiplet of fields Φ_a , a discrete action may be represented as

$$\Phi_a(x) \mapsto X_{ab} \Phi_b^\sharp(x'),$$

where x' is the spacetime-transformed point, $\sharp$ denotes any required complex conjugation or adjoint operation, and X may mix fields with identical gauge and Lorentz quantum numbers. The matrix X is not itself the symmetry. It is one presentation of the operation in a chosen field basis.

6.2 Charge conjugation

Charge conjugation exchanges a representation with its conjugate and maps particle sectors to antiparticle sectors. In a vectorlike theory such as QED, the action can be exact even though charged states are not individually fixed. In the chiral weak theory, conjugation exchanges left-handed and right-handed representation content that is not symmetrically present; C is therefore not a symmetry of the charged-current architecture. The correct statement is representation- and sector-indexed, not a theory-wide slogan.

A C-even operator need not imply full C invariance. Conversely, a non-invariant charged field may be part of a C-invariant action when its conjugate is included. The candidate record therefore distinguishes field transformation, action invariance and state-sector mapping.

6.3 Parity

Parity reverses spatial orientation. The QED and CP-even QCD interactions admit P in their fixed vectorlike targets. The weak charged current does not, because its V–A structure selects left-handed fermions. A nonzero strong $\bar{\theta}$ is P odd and T odd. The selected one-doublet scalar potential is P compatible, but that does not transport P to the full electroweak theory.

Intrinsic-parity phases, gamma-matrix conventions and coordinate orientation are presentations. The physical action is certified by its effect on states, local operators, angular correlations and amplitudes after quotienting those conventions.

6.4 Time reversal

Time reversal is antiunitary on physical rays. Its antilinearity complex-conjugates scalar coefficients and reverses the order appropriate to amplitudes. A displayed replacement $t \mapsto -t$, $\mathbf{p} \mapsto -\mathbf{p}$ is not a complete state-space operation. Direct T violation compares a transition with the correctly prepared and filtered time-reversed transition. An observed T-odd scalar triple product can be generated by absorptive phases and requires a separate ancestry certificate.

6.5 CP and generalized CP

CP combines charge conjugation and parity, but in a theory with repeated field representations the most general CP transformation can include a unitary map in flavor space. For left-handed and right-handed multiplets,

$$\psi_L(x) \mapsto X_L \mathcal{C} \overline{\psi_L}^T(x_P), \quad \psi_R(x) \mapsto X_R \mathcal{C} \overline{\psi_R}^T(x_P),$$

with target-compatible X_L, X_R . CP conservation is therefore an existence statement: there must exist a generalized CP map that preserves the complete action and selected vacuum. It is not the assertion that every matrix can be made entrywise real in one arbitrarily chosen basis.

6.6 CPT

CPT is kept as a distinct theorem object. SYM-2 separates the premise architecture, theorem consequence and antiunitary physical action. SYM-4 imports that packet and uses it to control CP–T relations. CPT exactness does not imply separate C, P, T or CP exactness, and failure of an individual discrete symmetry does not weaken the CPT theorem while its premises remain fixed [Maley, 2026r, Greaves and Thomas, 2014].

Discrete-action typing theorem

For a fixed target, each C/P/T/CP/CPT claim must identify: the acting operation, its state and field implementations, any flavor-unitary component, the acted-upon sector, the physical quotient, and the invariant object tested. An invariance or violation status stated without these data is incomplete.

6.7 Action versus fixed point

An invariant vacuum or CP eigenstate is a fixed point, not evidence that the operation acts trivially. Likewise, the absence of a nonzero asymmetry in one process does not establish that the action is exact in all sectors. The action, its stabilizers and its diagnostics are separate records.

Exact nonclaim

No claim is made that C, P, T or CP is primitive merely because it acts physically. Physical action and structural primitiveness are independent axes.

Active/Passive and Unitary/Antiunitary Noncollapse

7.1 Physical rays and phase skin

Pure states are rays,

$$Q_D^{\text{ray}} = (\mathcal{H} \setminus \{0\})/\mathbb{C}^\times.$$

Multiplying a state vector by a common nonzero phase does not change the physical ray. This fact is crucial in both CPT and large-gauge discussions. A phase attached to an implementation can be skin even when relative phases, superselection sectors or interference effects are physical. The classifier therefore asks what descends to the ray or observable algebra rather than treating any displayed phase as an action.

7.2 Active operations and passive representatives

A passive field-basis change reexpresses one operator algebra. An active discrete operation maps a physical state or observable to another physical state or observable. The same matrix notation can occur in both. Their statuses are fixed by the target and quotient, not typography.

Examples include:

- a gamma-matrix similarity transformation versus parity acting on a fermion state;
- a weak-basis rotation versus generalized CP acting on the Yukawa sector;
- a rephasing of external states versus a quotient-stable interference phase;
- a gauge transformation of the Higgs representative versus a physical CP action on gauge-invariant scalar operators;
- a relabeling of degenerate mass eigenvectors versus an operation changing an invariant transition probability.

SYM4-T07 – Active/passive discrete-action separation

An invertible convention, basis, coordinate, field-redefinition or common-phase change is presentation skin exactly when a validated witness preserves the target, physical ray, local operator algebra, interaction coefficients after simultaneous transformation, vacuum status, boundary/global data, continuation and failure profile. An active C/P/T/CP transformation is quotient-nontrivial when it changes a physical ray, observable, charge sector or selection rule after that presentation quotient.

7.3 Antiunitarity and complex conjugation

For antiunitary T ,

$$T(a|\psi\rangle + b|\phi\rangle) = a^*T|\psi\rangle + b^*T|\phi\rangle.$$

This antilinearity is not a convention. It is required by transition-probability preservation for the disconnected branch permitted by Wigner's theorem and controls the transformation of amplitudes. But complex conjugating a matrix entry in a chosen basis is not by itself the antiunitary physical action. The full operator, initial/final state interchange and target Hamiltonian must be specified.

SYM4-T08 – Antiunitary-T noncollapse

Physical T, naive reversal of displayed momenta and spins, complex conjugation of coefficients, reversal of a process arrow, and CPT-derived comparison are distinct atomic objects. A T claim is complete only when the antiunitary implementation and the initial/final channel map are fixed. Naive reversal can be a useful diagnostic convention but cannot occupy the physical T-action role.

7.4 Implementation phases and generalized transformations

C, P, T and CP operators are often defined up to intrinsic phases or field-family unitary maps. Such freedom is not automatically physical. The presentation witness must preserve the operation algebra, kinetic terms, interaction action, state rays and invariant diagnostics. If it changes an invariant CP-odd quantity or maps the target to a different theory, it is not skin.

7.5 Degenerate subspaces

Exact degeneracy enlarges the group of basis changes that preserve the mass operator. Angles and phases inside the degenerate subspace can become quotient-trivial. The project includes a controlled two-axis degeneracy target in which two compatible assignments are related by a nontrivial two-element lawful certificate-automorphism group. This verifies that the quotient removes presentation multiplicity without identifying the physical projector that defines the degenerate subspace.

Exact falsification route

A valid counterexample must exhibit a transformation classified as skin even though it changes a quotient-level amplitude, state, invariant or vacuum status, or a transformation classified as physical even though a complete invertible witness preserves the entire physical tuple.

Interaction-Sector and Process-Channel Restriction

8.1 Why there is no theory-wide Boolean bit

The Standard Model is a sum of interaction sectors and coupling structures. A discrete operation may preserve one projection and fail in another. QED can preserve P while the weak charged current violates it. The electroweak gauge kinetic terms can preserve CP while the quark Yukawa sector violates CP. QCD can preserve C while a nonzero $\bar{\theta}$ violates P, T and CP. A process asymmetry can be zero even when the underlying theory contains a nonzero CP-odd invariant.

The sector compiler therefore treats

$$\text{Status}(\delta) = \text{Status}(\delta; D, s, \text{channel}, \text{parameters}, V, \mathcal{A}, B, R)$$

as a typed function, not a bit.

8.2 Vectorlike sectors

In the fixed QED and CP-even QCD targets, left- and right-handed components occur in the vectorlike combination needed for C and P invariance of the interaction. The action can map charged states nontrivially while preserving the S-matrix. This projection does not include the chiral weak interaction, a CP-odd θ term, external chiral backgrounds or higher-dimension operators.

8.3 Chiral weak sectors

The charged-current interaction couples left-handed fermion doublets. Under P it would require a corresponding right-handed charged-current structure absent from the core target; under C it maps to conjugate chirality content that is not an equivalent representation of the same interaction. C and P are therefore violated by architecture before a CKM phase is considered. CP can nevertheless be preserved in a parameter limit when the Yukawa/mixing data admit a generalized CP map.

Neutral currents are also chiral. Their C and P status is not inherited from QED. Their tree-level flavor structure is separately constrained by the one-doublet Yukawa quotient; absence of tree-level flavor-changing neutral currents is not the same theorem as CP conservation [Maley, 2026s,n].

The closed target F-WEAK-NEUTRAL-CURRENT-TREE-3GEN fixes three generations, the one-doublet electroweak carrier, real tree-level Z couplings, the mass basis, and the same Yukawa quotient used by the charged-current target. At this locus the vector-axial current violates C and P, while its real coefficients make CP exact and the fixed CPT packet makes T exact. The unitary mass-basis transport keeps the family-universal neutral generators diagonal, so the absence of tree-level flavor-changing neutral currents is a derived flavor diagnostic rather than a new discrete symmetry. Independent phases of the nondegenerate mass eigenfields act by right multiplication on the diagonalization matrix and are quotiented by a complete mass-eigenstate rephasing solver; they do not create additional neutral-current physical configurations. Loop corrections, SMEFT operators, and nonunitary flavor structures are successor targets.

SYM4-T10 – Sector locality and no-uniform-bit theorem

No faithful record assigns one unqualified exact/violated value to C, P, T or CP for the complete Standard Model. The status is a function of interaction sector, channel, parameter region, vacuum, anomaly, boundary/global data and scale. The integrated answer is a qualified ledger, not a Boolean compression.

SYM4-T11 – Vectorlike/chiral sector theorem

Vectorlike electromagnetic and strong projections and chiral weak projections occupy different representation loci. A discrete-symmetry status proved in one does not transport to the other without an explicit sector inclusion and compatibility proof. Chiral C/P violation and phase-induced CP violation are distinct causal ancestries.

8.4 Process channels

Even inside one interaction sector, a direct observable statement is channel-specific. Initial-state preparation, final-state filtering, phase-space integration, spin analysis and detector matter effects can alter which combination of amplitudes is measured. The ledger therefore has separate fields for interaction sector and process channel.

For amplitudes

$$A_f = \sum_i a_i e^{i(\phi_i + \delta_i)},$$

where ϕ_i are weak phases and δ_i are CP-even strong or absorptive phases, rate asymmetries and T-odd correlations depend on different combinations. The existence of a theory-level CP-odd invariant does not guarantee a nonzero asymmetry in every channel; a nonzero T-odd correlation does not guarantee a T-odd weak phase.

8.5 Projection and recomposition

Each sector row is a projection of one Reference-complete physical fiber and is compiled by the integrated ledger assembly. Recomposition requires common field content, coupling convention, vacuum, anomaly ledger, CPT packet and parameter target. If two rows use different neutrino content or different θ conventions, they cannot be merged by name.

Exact falsification route

A proposed theory-wide sentence must be expandable into complete sector rows without contradiction or hidden target changes. If it cannot, the sentence is rejected as an overcompressed summary.

CPT Premise, Theorem, Implementation, and Relation Lattice

9.1 Imported packet

SYM-4 does not restate CPT as a one-line axiom. It imports the proposition-exact SYM-2 packet built on the selected Jost/Wightman target. That packet separates:

1. Minkowski geometry, orientation and spin-cover data;
2. positive metric, invariant vacuum and forward spectrum;
3. local field covariance, adjoint closure and weak local commutativity;
4. the analytic theorem interface;
5. the CPT theorem consequence;
6. the antiunitary physical implementation on states and gauge-invariant local fields;
7. charged-sector and formulation controls.

Only the canonical and normalized presentations certified by SYM-2 are treated as the same theorem target. Algebraic modular PCT, Lagrangian and curved-spacetime formulations remain distinct unless a bridge is supplied [Maley, 2026r, Greaves and Thomas, 2014].

9.2 Premise, theorem and action are distinct

The premise packet is a target structure. The theorem is a derived consequence. The antiunitary CPT operator is a physical action. Exactness and primitiveness therefore do not collapse. “Derived” does not mean optional, while “physical action” does not mean independently primitive.

SYM4-T09 – CPT relation lattice

Every CP–T inference is licensed only after the exact CPT premise and implementation packet is fixed. The ledger records separately:

1. direct CP status;
2. direct T status;
3. CPT theorem status;
4. the channel and state map used to transport one statement into another;
5. any target update invalidating that transport.

No T statement is generated by word substitution from CP alone.

9.3 Conditional interface rather than a physical violation fiber

The generic CPT relation is registered as a conditional theorem interface, not as a closed physical target preloaded with CP and T violation. CPT constrains the relation between statuses under its exact premises; it does not cause a violation. Concrete instantiations are supplied by physical fibers such as the nonzero-Jarlskog quark target, the

generic strong- $\bar{\theta}$ target, and the true transversity target. The interface is excluded from the closed-target orbit quantifier.

9.4 The CP–T relation

At a schematic amplitude level, CPT invariance constrains matrix elements between a process and the CPT-conjugate process. Under additional target and channel assumptions, CP violation implies T violation because CPT preserves the combined product. But the exact relation depends on state preparation, unstable-particle conventions, absorptive parts and whether one is comparing amplitudes, rates or effective Hamiltonian parameters.

The project therefore assigns one of four relation statuses:

Direct	T is tested through a transition and its properly defined time reverse.
CPT-linked	T status is inferred from CP using the fixed CPT packet and a verified channel map.
Diagnostic only	a T-odd or CP-odd observable is present but the full action comparison is not established.
Unavailable	the target violates or does not fix a CPT premise, or the channel map is incomplete.

9.5 CPT and unstable systems

Neutral mesons are open, decaying systems embedded in a larger unitary theory. Effective non-Hermitian Hamiltonians, flavor tags and decay filters require care. Direct T observations such as CPLEAR and BaBar construct experimentally accessible transitions related by time reversal rather than merely comparing a raw T-odd scalar [CPLEAR Collaboration, 1998, BaBar Collaboration, 2012]. SYM-4 records these as examples of the direct branch, not as premises of the structural classification.

9.6 Target updates

Lorentz violation, nonlocality, nonunitarity, curved/dynamical spacetime, thermal real-time targets and quantum-gravity regimes are not silently folded into the core. A claim that CPT fails or changes status in such a regime is a target update. The core theorem remains conditional on the SYM-2 packet.

SYM4-T20 – CPT-linked T theorem

Within a fixed target satisfying the imported CPT packet, a verified CP-violating physical relation transports to T violation only through the registered CPT and channel map. The result is target- and channel-specific. It does not replace a direct T test and does not apply after a CPT premise has changed.

Exact nonclaim

The manuscript sets no experimental limit on CPT violation and does not claim that the imported CPT formulation is the unique formulation for all QFTs.

Weak Basis, Rephasing, Degeneracy, and Generalized CP

10.1 Weak-basis transformations

Before Yukawa matrices are diagonalized, the kinetic and gauge terms admit independent unitary transformations of fermion multiplets. For the quark sector,

$$\begin{aligned} Q_L &\mapsto W_L Q_L, & u_R &\mapsto W_u u_R, & d_R &\mapsto W_d d_R, \\ Y_u &\mapsto W_L Y_u W_u^\dagger, & Y_d &\mapsto W_L Y_d W_d^\dagger. \end{aligned}$$

These transformations change matrix representatives while preserving the physical theory when all coupled objects are transformed coherently. They are basis transport, not a new physical flavor action on fixed invariant data.

After diagonalization, independent field rephasings and permutations remove additional matrix coordinates. If masses are degenerate, unitary rotations in the degenerate subspace enlarge the stabilizer. Any parameter count or CP criterion that ignores these stabilizers is target-incomplete.

SYM4-T12 – Basis and rephasing quotient

A quark or lepton mixing claim becomes physical only after quotienting simultaneous weak-basis changes, admissible rephasings, permutations and exact-degeneracy stabilizers. Matrix entries, Euler angles and displayed phases before that quotient are presentations. The quotient-stable spectra, mixing mismatch and invariant phase residues are physical response data.

10.2 Generalized CP

A generalized CP transformation can combine ordinary CP with unitary maps among fields carrying identical gauge quantum numbers. In the quark Yukawa sector the invariance conditions have the schematic form

$$\begin{aligned} X_L^\dagger Y_u X_u &= Y_u^*, \\ X_L^\dagger Y_d X_d &= Y_d^*, \end{aligned}$$

with unitary X_L, X_u, X_d compatible with the target. The existence of a solution is basis covariant: if the fields are weak-basis transformed, the X matrices transform with them.

SYM4-T13 – Generalized-CP existence criterion

A target is CP conserving at the coupling locus exactly when there exists a target-compatible generalized CP map preserving the complete action and all fixed backgrounds. The criterion is an existence statement on the quotient. It is not equivalent to the visual reality of one matrix representative.

10.3 Invariant certificates

Weak-basis invariant polynomials provide coordinate-independent tests. In the generic nondegenerate three-generation quark target, a canonical CP-odd invariant is proportional to

$$\text{Tr}([H_u, H_d]^3), \quad H_u = Y_u Y_u^\dagger, \quad H_d = Y_d Y_d^\dagger,$$

or equivalently to the determinant of the commutator. Its value factors into the Jarlskog invariant J and mass-squared differences [Jarlskog, 1985]. In that generic target, nonzero J with nondegenerate masses certifies CP violation.

The generic formula is not complete at exact degeneracy because the mass-difference factors vanish and the stabilizer changes. A complete target-relative certificate is a set of invariants sufficient for the declared degeneracy pattern, field content and operator locus. The project stores the generic Jarlskog witness as a diagnostic and the complete package as the normal-form occupant.

SYM4-T14 – Complete CP-odd invariant criterion

Physical CP violation is certified by a complete target-relative set of basis-invariant conditions or by an equivalent generalized-CP nonexistence proof. A complex coefficient or one invariant-looking expression is insufficient unless its completeness hypotheses are registered. In the generic nondegenerate three-generation quark target, the Jarlskog criterion is the canonical corollary.

10.4 Two generations and degeneracy

For two quark generations the unitary mixing matrix contains one physical angle and no irreducible Dirac-type phase after rephasing. Exact mass degeneracy similarly enlarges the stabilizer and can make a nominal phase or angle unphysical. These are not approximate statements. They are exact target-relative quotient results.

SYM4-T15 – Degeneracy and two-generation phase collapse

In the two-generation quark target, no irreducible CKM CP phase survives the weak-basis/rephasing quotient. In a target with exact mass degeneracy, rotations inside the degenerate subspace are skin unless an independent witness distinguishes them. The invariant certificate and parameter count must be recomputed for the enlarged stabilizer.

10.5 AASC status of mixing data

The corpus mixing papers classify a displayed phase as skin when removable, and a surviving phase as a tensor residue only when it is tied to a quotient-stable interference, asymmetry or holonomy witness [Maley, 2026t,u,v,w]. SYM-4 adopts that role discipline while retaining the mainstream QFT quotient as the physical certificate. It does not infer a numerical phase from structural standing.

Exact falsification route

A counterexample must preserve field content, mass degeneracies, the full weak-basis group and the complete invariant set while showing that the quotient removes a measurable CP distinction or retains a purely conventional phase.

Quark Yukawa Structure and Charged-Current CP

11.1 From Yukawa matrices to the CKM quotient

With one Higgs doublet, electroweak fixation produces quark mass matrices

$$M_u = \frac{v}{\sqrt{2}} Y_u, \quad M_d = \frac{v}{\sqrt{2}} Y_d.$$

Biunitary diagonalization gives

$$U_{uL}^\dagger M_u U_{uR} = D_u, \\ U_{dL}^\dagger M_d U_{dR} = D_d,$$

and the charged-current mismatch is

$$V_{\text{CKM}} = U_{uL}^\dagger U_{dL}.$$

The separate diagonalization matrices are basis-dependent. The mismatch class after the residual rephasing and permutation quotient is physical. The corpus Yukawa and mixing papers express the same distinction as carrier-compatible mass response versus quotient-stable mismatch [Maley, 2026x,t,u,n].

11.2 Parameter count

For N nondegenerate generations, a unitary mixing matrix has

$$\frac{N(N-1)}{2}$$

angles and

$$\frac{(N-1)(N-2)}{2}$$

Dirac-type phases after rephasing. Thus $N = 2$ admits one angle and no phase; $N = 3$ admits three angles and one phase. This is a quotient count, not a prediction of numerical values. In the observed target $N_g = 3$ is fixed boundary data [Maley, 2026g].

11.3 Generic three-generation CP violation

Kobayashi and Maskawa showed that three generations allow an irreducible CP phase in a renormalizable weak theory [Kobayashi and Maskawa, 1973]. Jarlskog supplied a basis-invariant test [Jarlskog, 1985]. For nondegenerate quark masses,

$$\det[H_u, H_d] \propto iJ \prod_{i < j} (m_{u_i}^2 - m_{u_j}^2) \prod_{k < \ell} (m_{d_k}^2 - m_{d_\ell}^2).$$

The invariant vanishes when CP is conserved, when the physical phase vanishes, or when a degeneracy enlarges the quotient. The factorization makes explicit why a displayed complex phase is not enough.

Quark CP normal-form theorem

In the fixed generic three-generation quark target, the physical CP record consists of the mass spectra, the weak-basis quotient, the CKM mismatch class, a complete CP-odd invariant certificate and the charged-

current action. Complex Yukawa entries, diagonalization matrices and rephasing conventions are skin. A nonzero complete certificate yields a quotient-nontrivial CP action and T violation through the separately imported CPT relation.

11.4 CP-conserving complex representatives

A basis may contain complex Y_u and Y_d even when a generalized CP transformation exists. Conversely, a basis in which one matrix is real does not establish full CP conservation if the second matrix and their relative orientation retain a nonzero invariant. SYM-4 includes a closed CP-conserving three-generation target to force this distinction.

The correct classification has three layers:

1. representative statement: are selected entries complex in this basis?
2. invariant statement: does the complete CP-odd set vanish?
3. action statement: does a generalized CP map preserve the complete target?

Only the latter two determine physical status.

11.5 Degeneracy and enhanced stabilizers

If two masses are exactly equal, the mass matrix admits additional unitary rotations. A phase that was physical in the generic target can become removable. The target therefore changes when degeneracy is imposed. The degenerate fiber has its own quotient and invariant package, rather than treating degeneracy as a small numerical perturbation of the generic theorem.

11.6 Neutral currents

The closed tree-level target F-WEAK-NEUTRAL-CURRENT-TREE-3GEN uses the same three-generation Yukawa quotient as the charged-current target. In the one-doublet renormalizable theory, the unitary transformations that diagonalize a fermion mass matrix leave each family-universal neutral generator flavor diagonal:

$$U_L^\dagger G_{L,R} U_L = G_{L,R}.$$

The CKM mismatch appears in the charged current because the up- and down-type left-handed fields share an $SU(2)_L$ doublet but are diagonalized differently. The executable target obtains off-diagonal generator norms below 10^{-15} and reports no tree-level flavor-changing neutral current. That result is a derived representation witness, not a C, P, T or CP action. The neutral current itself is chiral: C and P are violated at this locus, whereas real tree-level coefficients give exact CP and, through the fixed same-target CPT packet, exact T. Loop effects and higher-dimension flavor-changing neutral operators are explicit successor targets.

11.7 Numerical-status boundary

The manuscript does not compute the observed CKM angles, phase or Jarlskog value. Those are externally calibrated parameter data within the fixed structural target. Structural classification determines which combinations are physical and what would count as CP violation; it does not supply their numerical values.

Exact nonclaim

The existence of one irreducible phase for three generic generations is a parameter-count theorem. It is not a derivation that the phase is nonzero in nature. The observed nonzero value belongs to a separately frozen empirical target.

Strong CP: Axial Transport, Invariant Theta, Topology, and Vacuum Scope

12.1 The CP-odd QCD coordinate

The renormalizable QCD Lagrangian can contain

$$\mathcal{L}_\theta = \frac{g_s^2 \theta}{32\pi^2} G_{\mu\nu}^a \tilde{G}^{a\mu\nu}, \quad \tilde{G}^{a\mu\nu} = \frac{1}{2} \epsilon^{\mu\nu\rho\sigma} G_{\rho\sigma}^a.$$

The density is C even and P, T and CP odd. But θ alone is not the physical strong-CP coordinate when quark masses have complex phases.

Write the mass term as

$$-\bar{q}_R M_q q_L - \bar{q}_L M_q^\dagger q_R.$$

A common axial field redefinition changes the mass determinant phase and, through the anomalous Jacobian, shifts θ . In the adopted convention,

$$\theta \mapsto \theta + 2N_f \alpha, \quad \arg \det M_q \mapsto \arg \det M_q - 2N_f \alpha.$$

The invariant combination is

$$\bar{\theta} = \theta + \arg \det M_q.$$

The same result is worked explicitly in the supplied Standard Model closure source and is imported with the SYM-3 anomaly packet [Maley, 2026n,o, Fujikawa, 1979].

SYM4-T16 – Strong-CP invariant-combination theorem

For the fixed QCD mass, topology and anomaly convention, the strong-CP status depends on $\bar{\theta}$, not on θ or $\arg \det M_q$ separately. Axial field redefinitions relate presentations of the same invariant coordinate when their anomalous measure transport and mass transformation are included.

12.2 Axial redefinition is transport, not erasure

A chiral redefinition can move phase between the mass matrix and the topological term. It cannot remove a nonzero invariant merely by changing coordinates. The candidate record contains the source and target representatives, the measure Jacobian, the anomaly class, the mass determinant and the invariant endpoint.

SYM4-T17 – Axial-redefinition transport theorem

An axial redefinition is presentation-equivalent only when it transports θ , the complete mass matrix, the fermion measure, the anomaly convention, topology and downstream observables together. A redefinition that omits any of these is an incomplete certificate. A nonzero $\bar{\theta}$ cannot be canceled by representative change alone.

On the AASC role axis, the local phase convention is skin; the anomaly-aware transport is a licensed transport relation; $\bar{\theta}$ is a quotient-stable physical parameter datum; CP-odd observables are derived witnesses; and a mechanism adding new fields or dynamics is a target update.

12.3 Exactly massless quark target

If a quark is exactly massless, the corresponding axial rotation is not obstructed by a mass phase for that flavor and can be used to remove the physical strong phase. The removable object is the invariant parameter class $\bar{\theta}$; the discrete transformations are not “removable.” After the exact axial quotient is taken, the declared massless-quark target has exact C, P, T, and CP actions. This is an exact target-relative statement. It requires exact masslessness and the corresponding operator content; an approximately light quark does not supply the same theorem. SYM-4 therefore treats the massless-quark target as a separate closed fiber.

12.4 Generic strong-CP status

For generic nonzero quark masses, a generic nonzero $\bar{\theta}$ violates P, T and CP while preserving C. The manuscript classifies that status; it does not predict the value of $\bar{\theta}$ or explain its observed smallness. Electric dipole moments and hadronic matrix elements are downstream observables requiring separate QCD and experimental certificates.

12.5 Topology and faithful global form

The physical meaning and periodicity of θ depend on the admitted bundle sectors, faithful global gauge group and extended operator category. A local density and perturbative anomaly calculation do not by themselves fix those global data. SYM-1 and SYM-3 parameterize the dependence, while full line/monopole and global-form closure is assigned to SYM-9 [Maley, 2026y,o, Tong, 2017, Davighi et al., 2020].

12.6 Special values and vacuum branching

At special values such as $\theta = \pi$, CP can be compatible with the action modulo periodicity while the vacuum may realize CP nontrivially or spontaneously, depending on masses, global form and branch structure. A statement such as “CP is restored at $\theta = \pi$ ” is therefore incomplete unless periodicity and vacuum realization are fixed.

SYM4-T18 – Special- θ routing theorem

A special- θ CP claim enters core closure only when the faithful global form, topological sectors, periodicity, quark-mass target and selected vacuum branch are fixed. Otherwise it is an explicit SYM-9/vacuum frontier. It may not be imported into the generic strong-CP row by notation alone.

12.7 Strong-CP relation lattice

The project distinguishes:

1. raw θ representative;
2. quark-mass determinant phase;
3. axial field redefinition;
4. anomalous measure transport;
5. invariant $\bar{\theta}$;
6. C action status, which remains exact across the closed strong-CP fibers;
7. P/T/CP action status;
8. topological periodicity;

9. vacuum branch;
10. observable witness;
11. added-sector mechanisms.

No two are merged merely because they appear in one conventional formula.

Exact nonclaim

SYM-4 does not solve the strong-CP problem, select an axion or Peccei–Quinn mechanism, predict a neutron electric dipole moment, or close special- θ global-form and vacuum questions.

Time Reversal, Direct Tests, and Observable Ancestry

13.1 Antiunitary T and kinematic reversal

Physical time reversal is an antiunitary operation on the state space. It is not identical to reversing displayed momenta and spins while leaving initial and final states in place. That latter convention, often called naive T, is useful for constructing pseudoscalar angular correlations but does not by itself compare a process with its time-reversed transition.

13.2 Concrete transversity target

The maximum-stress target fixes a spin-zero parent decaying to two vector final states. In the transversity basis,

$$A_h = a_h e^{i(\delta_h + \phi_h)}, \quad \bar{A}_h = \eta_h a_h e^{i(\delta_h - \phi_h)}, \quad h \in \{0, \parallel, \perp\},$$

where δ_h are CP-even absorptive phases, ϕ_h are weak phases, and $\eta_0 = \eta_{\parallel} = +1$, $\eta_{\perp} = -1$ in the declared convention. The raw T-odd transversity correlations include

$$T_0 = \text{Im}(A_{\perp} A_0^*), \quad T_{\parallel} = \text{Im}(A_{\perp} A_{\parallel}^*),$$

and the corresponding CP-conjugate quantities. True and fake combinations are formed from the appropriately convention-adjusted sum and difference. Their precise sign convention is frozen in the target and constructor record; the invariant distinction is that the true combination requires a weak-phase difference, while the fake combination can survive from absorptive phases alone [Datta and London, 2004].

The executed fake target sets all ϕ_h equal and selects unequal δ_h . It produces a nonzero raw triple product and a nonzero fake combination but a vanishing true CP/T-odd combination. The contrasting true target contains a nonzero weak-phase difference and produces a nonzero true combination. These are formula-level outputs, not descriptive labels.

SYM4-T19 – T-odd-correlation noncollapse

A nonzero observable odd under naive momentum/spin reversal is not sufficient for fundamental T violation. Its certificate must specify the amplitude basis, absorptive and weak phases, CP-conjugate channel, antiunitary-T action, and inverse-transition relation. The raw correlation, its true/fake decomposition, and the physical T action receive separate records.

13.3 Direct T tests

A direct T test compares the probability for $i \rightarrow f$ with the probability for the time-reversed transition $Tf \rightarrow Ti$, with preparation and filtering matched. Neutral-meson experiments provide examples in which experimentally accessible transitions were organized for direct T comparisons [CPLEAR Collaboration, 1998, BaBar Collaboration, 2012]. The manuscript uses these experiments to fix the logical distinction, not to import one measured number as an ancestor of the structural theorem.

13.4 CPT-linked inference

When the exact SYM-2 CPT premise and implementation packet is fixed, a physical CP-violating relation implies T violation at the corresponding same target. This is a theorem transport, not an identification of the two operations and not a substitute for a direct inverse-transition test.

SYM4-T20 – CPT-linked T theorem

A CP-to-T inference is licensed only when the exact CPT premise packet, target, channel, and physical implementation are preserved. Direct T comparison, CPT-linked inference, and a kinematic T-odd diagnostic are distinct theorem objects even when they agree in one target.

13.5 External fields, matter effects, and boundaries

A magnetic field, material environment, asymmetric boundary, or matter potential can change a P or T comparison. Such data are target coordinates. Neutrino propagation through matter, for example, can generate neutrino–antineutrino probability differences that must be separated from intrinsic mixing phases in an experimental target.

Exact falsification route

A claim of fundamental T violation based only on a triple-product sign, with no amplitude ancestry or inverse-transition certificate, is rejected. Conversely, a direct transition comparison cannot be demoted merely because a CPT-linked route is also available.

Higgs Carrier, Scalar CP, and Vacuum Realization

14.1 The one-doublet scalar target

The renormalizable one-Higgs-doublet Standard Model scalar potential is

$$V(H) = -\mu^2 H^\dagger H + \lambda (H^\dagger H)^2,$$

with real μ^2 and λ by Hermiticity. A vacuum representative can be written

$$\langle H \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v e^{i\xi} \end{pmatrix}.$$

In the one-doublet gauge target, the phase ξ can be removed by a gauge transformation; it is not an independent gauge-invariant order parameter. The gauge-invariant scalar spectrum contains one neutral Higgs-like excitation and no independent scalar-sector spontaneous-CP phase.

At the isolated scalar locus the conjugation map $H(x) \mapsto H^*(x)$ preserves the real potential and the selected real gauge-invariant vacuum orbit. The machine ledger therefore records exact C, P, T and CP for this scalar projection. These rows are locus restricted: they do not restore C or P in the chiral electroweak fermion sector.

This statement is deliberately restricted. It does not say that the complete Standard Model conserves CP: quark and neutrino Yukawa sectors and strong $\bar{\theta}$ have separate records. It does not say that extended scalar sectors cannot violate CP: two-doublet and other models are successor targets, historically including spontaneous-CP constructions [Lee, 1973].

14.2 Gauge symmetry and vacuum language

Local gauge redundancy is not assigned a gauge-variant order parameter as a physical spontaneous-breaking witness [Elitzur, 1975, Maley, 2026y]. Gauge-fixed vev and Goldstone coordinates remain useful perturbative representatives. Physical statements are made through gauge-invariant observables, residual electromagnetic charge, pole data and the carrier/fixation architecture.

14.3 Higgs Arc imports

The Higgs Structural Endpoint Arc supplies a precise standing map:

1. $C_H = H_{\text{br/fix}}$ is the bridge/fixation carrier required by electroweak mass response [Maley, 2026h];
2. Higgs identity is unique occupation of C_H , not a primitive vev, scalar coordinate or potential minimum [Maley, 2026i];
3. Higgs-associated scales are carrier-indexed rather than independent primitive grounds [Maley, 2026j];
4. vacuum location, lower branches, metastability and thermal routes are route statuses; route availability does not imply standing closure [Maley, 2026k];
5. extra scalar sectors are classified by their relation to C_H and become enrichment or target update unless they preserve and reassemble the same carrier role [Maley, 2026l];
6. no residual same-scope primitive Higgs attribute survives outside the closed endpoint architecture [Maley, 2026m].

The CP analysis inherits these results without turning them into substitute QFT premises. The physical no-spontaneous-CP result comes from the one-doublet gauge-invariant target. The AASC result determines whether the vev phase, vacuum route, extension or scalar representative can ground identity or standing.

SYM4-T21 – Explicit/spontaneous CP separation

A CP violation claim is explicit when no target-compatible generalized CP map preserves the coupling data. It is spontaneous when the action admits such a map but the selected gauge-invariant vacuum does not. Coupling and vacuum certificates are independent. A gauge-fixed phase is not a spontaneous order parameter without a gauge-invariant witness.

SYM4-T22 – One-Higgs-doublet scalar/vacuum theorem

In the fixed renormalizable one-doublet Standard Model target, the scalar potential and gauge-invariant vacuum contain no independent spontaneous-CP phase. The Higgs field coordinate, vev phase and potential chart are representatives licensed by the bridge/fixation carrier. A scalar-sector claim requiring an additional invariant phase, order parameter or Higgs identity is a target update or an incomplete certificate.

14.4 Extended scalar targets

A second Higgs doublet introduces new gauge-invariant relative phases and a richer potential. Singlets, composite sectors, portals, supersymmetric fields and pNGB realizations can likewise alter the scalar and vacuum ledger. The Higgs extension classifier does not forbid them. It requires them to state whether they preserve, enrich, split, replace or update the carrier architecture.

SYM-4 therefore routes an extended-scalar CP claim to U-HIGGS-EXTENDED. The successor target must provide:

- complete scalar field content and gauge representations;
- the potential and generalized CP maps;
- basis-invariant explicit-CP tests;
- a gauge-invariant vacuum certificate;
- mixing and mass-eigenstate projections;
- anomaly and boundary/global-form data where relevant;
- a carrier relation to C_H .

14.5 Vacuum-route status

Metastability, lower branches, tunneling lifetimes and thermal histories can be physically important without being primitive CP or Higgs grounds. If a thermal path or electroweak-baryogenesis route involves CP-violating sources, those sources require their own coupling and transport certificates. Route availability does not establish that the one-doublet core contains spontaneous CP.

Exact nonclaim

The manuscript neither computes the electroweak vacuum lifetime nor assesses the phenomenology of extended Higgs sectors. It classifies their target and standing status relative to the fixed core.

Neutrino Targets, PMNS Quotients, and Leptonic CP

15.1 Three noninterchangeable targets

Neutrino CP status is not defined until the field content and mass type are fixed. SYM-4 keeps three targets separate.

Minimal massless target. The minimal renormalizable Standard Model contains no right-handed neutrinos and no neutrino mass term. The lepton-flavor rotations that would define a PMNS matrix are not fixed by distinct neutrino mass eigenstates. A displayed PMNS matrix is therefore presentation, not physical mixing data. The machine certificate normalizes the former null sentinel to the identity as a canonical representative and then quotients the complete right $U(3)$ orbit, so every explicit unitary display belongs to the same massless-neutrino presentation class. The absence of a physical PMNS phase does not remove the discrete operations: in the declared real-coupling massless-neutrino target the CP and T actions are exact, while the PMNS coordinate and leptonic CP phase are absent only as physical phase data after the enlarged flavor quotient.

Dirac target. Right-handed neutrinos and Dirac masses produce a mismatch between charged-lepton and neutrino mass bases. After the Dirac rephasing quotient, the three-generation PMNS matrix has three angles and one Dirac-type CP phase.

Majorana target. A Majorana mass term restricts neutrino rephasings. For three nondegenerate Majorana neutrinos the mixing matrix contains the Dirac-type phase plus two additional Majorana phase coordinates. Their physical operator locus differs from ordinary oscillation.

SYM4-T23 – Neutrino target bifurcation

Minimal massless, Dirac and Majorana neutrino constructions are different targets with different quotients and phase counts. No PMNS, lepton-number or CP statement may be transported among them without an explicit target map.

15.2 Boundary reconstruction and the oscillation kernel

The supplied neutrino corpus treats oscillation as a source–propagation–detection reconstruction. The fixed structural object contains boundary flavor projectors, the propagation operator and the rephasing-invariant witness ledger [Maley, 2026q,p,z]. In vacuum,

$$\mathcal{A}_{\alpha \rightarrow \beta}(L) = \sum_i U_{\beta i} \exp\left(-i \frac{m_i^2 L}{2E}\right) U_{\alpha i}^*.$$

The probability depends on quartets

$$Q_{\alpha\beta}^{ij} = U_{\alpha i} U_{\beta i}^* U_{\alpha j}^* U_{\beta j},$$

which are invariant under charged-lepton and Dirac-neutrino rephasings. The imaginary parts are CP-odd oscillation witnesses.

SYM4-T24 – Dirac oscillation CP theorem

In the fixed Dirac-neutrino target, intrinsic vacuum oscillation CP violation is certified by nonzero imaginary rephasing-invariant PMNS quartets together with nondegenerate propagation phases. A displayed parameter δ_{CP} has physical standing only as a coordinate of that quotient-stable invariant content. Matter-induced asymmetry is a separate target coordinate.

15.3 Majorana phases and operator locus

A convenient Majorana parameterization is

$$U_{\text{PMNS}} = V(\theta_{12}, \theta_{13}, \theta_{23}, \delta) \text{diag}(1, e^{i\alpha_{21}/2}, e^{i\alpha_{31}/2}).$$

The rightmost phases cancel from the oscillation quartets because each mass-eigenstate column appears with its complex conjugate. They can enter lepton-number-violating amplitudes, such as the effective mass combination

$$m_{ee} = \left| \sum_i U_{ei}^2 m_i \right|,$$

where the square rather than modulus-square retains Majorana phase information. This does not make a measured or bounded m_{ee} a direct numerical phase readout without the masses, nuclear matrix elements and process certificate.

SYM4-T25 – Majorana phase-locus theorem

Majorana phases survive the restricted rephasing quotient of a Majorana target but are absent from ordinary oscillation probabilities. Their physical standing requires a lepton-number-violating operator or amplitude certificate. Dirac oscillation CP and Majorana phase loci are separate records.

15.4 Ordering, octant and phase orientation

The AOT2 source imposes a prediction-lock discipline: a structural mixing object does not by itself select mass ordering, octant, a numerical CP phase, absolute scale or Dirac/Majorana status. A branch readout requires an invariant projection from the orientation-neutral witness set. Current fit preferences cannot be imported upstream as derivation inputs [Maley, 2026p].

SYM-4 uses this result to separate:

- existence and count of phase coordinates;
- nonzero invariant CP standing;
- numerical phase calibration;
- branch or ordering readout;
- matter/background effects;
- Majorana operator-locus evidence.

15.5 Matter effects

Propagation through matter adds a flavor-dependent potential and breaks the simple vacuum CP comparison. Neutrino and antineutrino probabilities can differ even for a CP-conserving vacuum mixing matrix. An exper-

imental CP claim must therefore state the density profile, baseline, energy range and analysis used to separate intrinsic and matter-induced effects. These are not part of the structural vacuum fiber.

15.6 Generation count

The observed three-generation neutrino and charged-lepton matrices are defined in a target with $N_g = 3$ fixed as boundary data. SYM-4 does not derive this cardinality. It uses the generation-count theorem to prevent a parameter-count statement from being misreported as a generation prediction [Maley, 2026g].

Exact nonclaim

No neutrino mass ordering, octant, absolute mass scale, numerical Dirac phase or Majorana phase is predicted. The manuscript classifies the quotient and witness conditions under which such quantities are physical or empirically readable.

Discrete Anomalies, Global Form, and Generalized-Operator Frontiers

16.1 Anomaly import discipline

SYM-3 establishes that “anomaly” is not one status. A dynamical gauge anomaly is inconsistency; an anomalous global current can remain a meaningful physical action; a ’t Hooft anomaly obstructs gauging and constrains infrared realization; a global determinant phase can survive local cancellation; and a counterterm can redistribute local representatives without removing an invariant obstruction [Maley, 2026o].

Discrete-symmetry analysis inherits that taxonomy. A finite transformation can have a global anomaly even when no infinitesimal current exists. The target must specify the finite group, fermionic or bosonic nature, tangential structure, faithful action, background bundles, allowed counterterms and operator category. The word “mod N ” is not a proof record.

SYM4-T26 – Discrete-anomaly import theorem

A C/P/T/CP-related anomaly row imports its anomaly class, target, global form, tangential structure and consequence unchanged from SYM-3 or a registered successor. SYM-4 may classify the resulting discrete action, obstruction and sector status, but it may not convert a background ’t Hooft anomaly into gauge inconsistency or infer global freedom from local cancellation.

16.2 Spacetime-reflection anomalies

P, T and CP can involve orientation reversal and may require pin, spin or other tangential structures in a global anomaly analysis. Such questions are outside a purely local flat-space Lagrangian ledger. The core SYM-4 target records the interface and does not claim a complete bordism classification of every discrete spacetime symmetry.

16.3 Faithful global form and strong theta

Different faithful Standard Model gauge groups

$$G_{\text{SM}} = \frac{SU(3)_c \times SU(2)_L \times U(1)_Y}{\Gamma}$$

can share the same local Lie algebra and perturbative particle content while differing in bundles, line operators, monopoles, theta periodicities and generalized symmetries [Tong, 2017, Davighi et al., 2020, Maley, 2026y,o]. A CP claim at a special theta value can therefore depend on data not fixed by the local theory.

16.4 Higher-form, higher-group and non-invertible structures

A zero-form flavor or discrete symmetry can mix with one-form data in a higher group; non-invertible defects can replace a naive broken group action in massless limits. These structures act on lines, surfaces or defects rather than only on local fields. The SYM-4 local ledger cannot close them by adding names to a point-particle table.

SYM4-T27 – Global-form and generalized-operator frontier theorem

A discrete-symmetry claim whose truth depends on faithful global form, genuine lines or monopoles, higher-group background transformations, non-invertible defects or generalized anomaly data is an explicit SYM-9 frontier unless the complete operator-category target is supplied. Such a row is not counted as a failure of the sharp-local SYM-4 normal form and is not silently imported into it.

16.5 Boundary and inflow

A discrete anomaly can be realized by inflow from a higher-dimensional bulk. The combined bulk–boundary system is a different target from an anomalous boundary theory alone. Inflow is not retrospective repair. It changes the fixed system whose partition function is required to be invariant.

16.6 Frontier record

The canonical ledger includes X-DISCRETE-ANOMALY and X-THETA-PI-GLOBALFORM. Their presence matters: the finite theorem closes only Reference-complete fibers. The frontier rows state exactly which data are missing and which successor paper owns them.

Exact nonclaim

No claim is made that every discrete anomaly relevant to the Standard Model has been inventoried. The complete global-form and generalized-symmetry burden remains SYM-9.

Part III

Canonical Standard Model Discrete-Symmetry Ledger

Charge-Conjugation Ledger

17.1 Reader-facing result

Charge conjugation is exact in the fixed vectorlike electromagnetic projection and in the conventional QCD projection, including the fact that the $G\widetilde{G}$ density is C even. It is not a symmetry of the chiral charged-current weak architecture and is generally not a symmetry of the neutral-current weak interaction. The one-doublet scalar potential is C compatible at its own locus, but that local statement does not determine the Yukawa or full electroweak status.

The central distinction is representation response. Charge conjugation exchanges a representation with its conjugate. In a vectorlike sector the conjugate response is present in the same interaction architecture. In a chiral sector it need not be. This is a structural C violation independent of a CKM or PMNS phase. The closed tree-level neutral-current target therefore records C violation even though CP and T are exact there. Conversely, the isolated real one-doublet scalar potential and vacuum orbit admit the exact scalar conjugation $H \mapsto H^*$; that scalar-locus result does not determine the full electroweak fermion sector.

17.2 Action, current and diagnostic

A C transformation can act nontrivially on states while leaving the dynamics invariant. Particle and antiparticle states are not identified in the physical quotient merely because C is exact. Currents, selection rules and decay asymmetries are diagnostics; they are not the action itself.

17.3 Canonical C rows

Table 17.1: C sector ledger.

Record	Target	Sector/channel	Physical status	Quotient	Invariant certificate	CPT dependency	Role
SYM4-L-001	F-QED-VECTORLIKE-REALMASS	QED	exact	quotient-nontrivial physical action when defined; target/sector indexed	fixed CPT premise and same-target CP/T relation	not used or separately declared	PhysicalAction
SYM4-L-005	F-WEAK-CHIRAL-1GEN	weak charged current	violated	quotient-nontrivial physical action when defined; target/sector indexed	fixed CPT premise and same-target CP/T relation	not used or separately declared	PhysicalAction; DerivedWitness
SYM4-L-009	F-WEAK-NEUTRAL-CURRENT-TREE-3GEN	weak neutral current	violated	quotient-nontrivial physical action when defined; target/sector indexed	tree-level neutral current; FCNC=False; offdiag=(2.2e-16,4.05e-17)	not used or separately declared	PhysicalAction; DerivedWitness
SYM4-L-013	F-CKM-GENERIC-3GEN	quark charged current	violated	quotient-nontrivial physical action when defined; target/sector indexed	J=2.9295157e-05; det[Hu,Hd]=237448.08	Im not used or separately declared	PhysicalAction; DerivedWitness
SYM4-L-017	F-CKM-CPZERO-3GEN	quark charged current	violated	quotient-nontrivial physical action when defined; target/sector indexed	J=-0; Im det[Hu,Hd]=0	not used or separately declared	PhysicalAction; DerivedWitness
SYM4-L-021	F-CKM-2GEN	quark charged current	violated	quotient-nontrivial physical action when defined; target/sector indexed	J=0; Im det[Hu,Hd]=0	not used or separately declared	PhysicalAction; DerivedWitness
SYM4-L-025	F-CKM-DEGENERATE-UC	quark charged current	violated	quotient-nontrivial physical action when defined; target/sector indexed	J=0; Im det[Hu,Hd]=0	not used or separately declared	PhysicalAction; DerivedWitness
SYM4-L-029	F-DEGENERATE-2D-AUTOMORPHISM	degenerate flavor control	not_applicable	quotient-nontrivial physical action when defined; target/sector indexed	fixed CPT premise and same-target CP/T relation	not used or separately declared	PhysicalAction
SYM4-L-033	F-STRONGCP-GENERIC	QCD theta	exact	quotient-nontrivial physical action when defined; target/sector indexed	bar-theta=0.8500000000000125; periodicity=6.283185307179586; uum=invariant	pe-vac not used or separately declared	PhysicalAction
SYM4-L-037	F-STRONGCP-THETA0	QCD theta	exact	quotient-nontrivial physical action when defined; target/sector indexed	bar-theta=1.2434497875801753e-14; periodicity=6.283185307179586; uum=invariant	vac not used or separately declared	PhysicalAction
SYM4-L-041	F-STRONGCP-THETAPI-INVARIANT	QCD theta	exact	quotient-nontrivial physical action when defined; target/sector indexed	bar-theta=-3.1415926535897807; periodicity=6.283185307179586; uum=invariant	pe-vac not used or separately declared	PhysicalAction
SYM4-L-045	F-STRONGCP-THETAPI-BROKEN	QCD theta	exact	quotient-nontrivial physical action when defined; target/sector indexed	bar-theta=-3.1415926535897807; periodicity=6.283185307179586; uum=spontaneously_broken	pe-vac not used or separately declared	PhysicalAction
SYM4-L-049	F-STRONGCP-MASSLESS-U	QCD theta	exact	quotient-nontrivial physical action when defined; target/sector indexed	bar-theta=None; periodicity=6.283185307179586; uum=invariant	periodic-vac not used or separately declared	PhysicalAction
SYM4-L-053	F-HIGGS-ONE-DOUBLET	one-Higgs-doublet scalar	exact	quotient-nontrivial physical action when defined; target/sector indexed	real one-doublet potential; gauge-invariant vacuum orbit	gauge not used or separately declared	PhysicalAction
SYM4-L-057	F-NU-MASSLESS	leptonic charged current	violated	quotient-nontrivial physical action when defined; target/sector indexed	Josc=0; Majorana invariants=[]	not used or separately declared	PhysicalAction; DerivedWitness
SYM4-L-061	F-NU-DIRAC	leptonic charged current	violated	quotient-nontrivial physical action when defined; target/sector indexed	Josc=0.029977694; Majorana invariants=[]	Majorana invari not used or separately declared	PhysicalAction; DerivedWitness
SYM4-L-065	F-NU-MAJORANA-WEINBERG	leptonic charged current	violated	quotient-nontrivial physical action when defined; target/sector indexed	Josc=0.024129829; Majorana invariants=[-0.07955232138350578, 0.01218800169533607]	Majorana invariants not used or separately declared	PhysicalAction; DerivedWitness
SYM4-L-075	A-SYM4-INTEGRATED-LEDGER	integrated ledger	sector-indexed	quotient-nontrivial physical action when defined; target/sector indexed	target-relative transformation and sector certificate	not used or separately declared	PhysicalAction

17.4 Blocked inferences

The ledger blocks the following shortcuts:

- C invariance of QED does not imply C invariance of the full electroweak theory;
- absence of a C-odd observable in one channel does not establish sector exactness;
- a C-even strong- θ density does not make the complete strong sector CP even;
- a neutral scalar does not by itself establish C or CP status for its Yukawa couplings;
- charge conjugation of a field representation is not the same as a weak-basis complex conjugation.

C ledger endpoint

The integrated answer is: C is a physical action with exact vectorlike projections and chiral weak violations. Its status is sector-local and is not controlled by the same invariant data as CP violation.

Parity Ledger

18.1 Reader-facing result

Parity is exact in the fixed QED projection and in QCD when the strong CP-odd invariant is in a P-even class. It is violated by the V–A charged-current interaction and generally by weak neutral currents. A generic nonzero $\bar{\theta}$ violates parity in QCD. The one-doublet scalar potential and its gauge-invariant vacuum are parity compatible at the scalar locus, while the complete Standard Model is not.

18.2 Architecture versus parameter violation

Weak P violation arises from chiral representation architecture. Strong-sector P violation from $\bar{\theta}$ is parameter-dependent. These causes must not be collapsed. Removing the CKM phase does not restore parity to the charged-current interaction. Setting $\bar{\theta} = 0$ does not alter the weak chiral architecture.

18.3 Vacuum and boundary qualifications

A parity statement can depend on external fields, material boundaries or an orientation-sensitive background. Those are target data. A gauge-fixed Higgs representative is not a parity order parameter unless a gauge-invariant operator changes.

18.4 Canonical P rows

Table 18.1: P sector ledger.

Record	Target	Sector/channel	Physical status	Quotient	Invariant certificate	CPT dependency	Role
SYM4-L-003	F-QED-VECTORLIKE-REALMASS	QED	exact	quotient-nontrivial physical action when defined; target/sector indexed	fixed CPT premise and same-target CP/T relation	not used or separately declared	PhysicalAction
SYM4-L-007	F-WEAK-CHIRAL-1GEN	weak charged current	violated	quotient-nontrivial physical action when defined; target/sector indexed	fixed CPT premise and same-target CP/T relation	not used or separately declared	PhysicalAction; DerivedWitness
SYM4-L-011	F-WEAK-NEUTRAL-CURRENT-TREE-3GEN	weak neutral current	violated	quotient-nontrivial physical action when defined; target/sector indexed	tree-level neutral current; FCNC=False; offdiag=(2.2e-16,4.05e-17)	not used or separately declared	PhysicalAction; DerivedWitness
SYM4-L-015	F-CKM-GENERIC-3GEN	quark charged current	violated	quotient-nontrivial physical action when defined; target/sector indexed	J=2.9295157e-05; det[Hu,Hd]=237448.08	Im not used or separately declared	PhysicalAction; DerivedWitness
SYM4-L-019	F-CKM-CPZERO-3GEN	quark charged current	violated	quotient-nontrivial physical action when defined; target/sector indexed	J=-0; Im det[Hu,Hd]=0	not used or separately declared	PhysicalAction; DerivedWitness
SYM4-L-023	F-CKM-2GEN	quark charged current	violated	quotient-nontrivial physical action when defined; target/sector indexed	J=0; Im det[Hu,Hd]=0	not used or separately declared	PhysicalAction; DerivedWitness
SYM4-L-027	F-CKM-DEGENERATE-UC	quark charged current	violated	quotient-nontrivial physical action when defined; target/sector indexed	J=0; Im det[Hu,Hd]=0	not used or separately declared	PhysicalAction; DerivedWitness
SYM4-L-031	F-DEGENERATE-2D-AUTOMORPHISM	degenerate flavor control	not_applicable	quotient-nontrivial physical action when defined; target/sector indexed	fixed CPT premise and same-target CP/T relation	not used or separately declared	PhysicalAction
SYM4-L-035	F-STRONGCP-GENERIC	QCD theta	violated	quotient-nontrivial physical action when defined; target/sector indexed	bar-theta=0.8500000000000125; periodicity=6.283185307179586; uum=invariant	pe-vac not used or separately declared	PhysicalAction; DerivedWitness
SYM4-L-039	F-STRONGCP-THETA0	QCD theta	exact	quotient-nontrivial physical action when defined; target/sector indexed	bar-theta=1.2434497875801753e-14; periodicity=6.283185307179586; uum=invariant	vac not used or separately declared	PhysicalAction
SYM4-L-043	F-STRONGCP-THETAPI-INVARIANT	QCD theta	exact	quotient-nontrivial physical action when defined; target/sector indexed	bar-theta=-3.1415926535897807; periodicity=6.283185307179586; uum=invariant	pe-vac not used or separately declared	PhysicalAction
SYM4-L-047	F-STRONGCP-THETAPI-BROKEN	QCD theta	spontaneously_broken	quotient-nontrivial physical action when defined; target/sector indexed	bar-theta=-3.1415926535897807; periodicity=6.283185307179586; uum=spontaneously_broken	pe-vac not used or separately declared	PhysicalAction; DerivedWitness
SYM4-L-051	F-STRONGCP-MASSLESS-U	QCD theta	exact	quotient-nontrivial physical action when defined; target/sector indexed	bar-theta=None; periodicity=6.283185307179586; uum=invariant	periodic-vac not used or separately declared	PhysicalAction
SYM4-L-055	F-HIGGS-ONE-DOUBLET	one-Higgs-doublet scalar	exact	quotient-nontrivial physical action when defined; target/sector indexed	real one-doublet potential; gauge-invariant vacuum orbit	gauge not used or separately declared	PhysicalAction
SYM4-L-059	F-NU-MASSLESS	leptonic charged current	violated	quotient-nontrivial physical action when defined; target/sector indexed	Josc=0; Majorana invariants=[]	not used or separately declared	PhysicalAction; DerivedWitness
SYM4-L-063	F-NU-DIRAC	leptonic charged current	violated	quotient-nontrivial physical action when defined; target/sector indexed	Josc=0.029977694; Majorana invariants=[]	Majorana invari not used or separately declared	PhysicalAction; DerivedWitness
SYM4-L-067	F-NU-MAJORANA-WEINBERG	leptonic charged current	violated	quotient-nontrivial physical action when defined; target/sector indexed	Josc=0.024129829; Majorana invariants=[-0.07955232138350578, 0.01218800169533607]	Majorana invariants not used or separately declared	PhysicalAction; DerivedWitness
SYM4-L-078	A-SYM4-INTEGRATED-LEDGER	integrated ledger	sector-indexed	quotient-nontrivial physical action when defined; target/sector indexed	target-relative transformation and sector certificate	not used or separately declared	PhysicalAction

P ledger endpoint

Parity has exact vectorlike projections, a parameter-indexed strong-sector status and architecture-level weak violations. No unqualified sentence “the Standard Model violates parity” is accepted without this decomposition.

Time-Reversal Ledger

19.1 Reader-facing result

Time reversal is an antiunitary physical action. In fixed QED and CP-even QCD targets it is exact. A generic nonzero $\bar{\theta}$ violates T. The quark charged-current sector violates T when the complete CP-odd invariant certificate is nonzero and the CPT packet is fixed. Direct T tests and CPT-linked inferences are distinct. T-odd correlations with absorptive phases are diagnostics, not sufficient action-level proofs.

19.2 Four statuses under one label

The table distinguishes:

1. exact antiunitary action;
2. direct T violation through inverse-transition comparison;
3. CPT-linked T violation inferred from CP;
4. T-sensitive diagnostic with unresolved or absorptive ancestry.

This classification prevents naive vector reversal from masquerading as the physical operation.

19.3 Canonical T rows

Table 19.1: T sector ledger.

Record	Target	Sector/channel	Physical status	Quotient	Invariant certificate	CPT dependency	Role
SYM4-L-004	F-QED-VECTORLIKE-REALMASS	QED	exact	quotient-nontrivial physical action when defined; target/sector indexed	fixed CPT premise and same-target CP/ T relation	P-SYM2-CPT	PhysicalAction
SYM4-L-008	F-WEAK-CHIRAL-1GEN	weak charged current	exact	quotient-nontrivial physical action when defined; target/sector indexed	fixed CPT premise and same-target CP/ T relation	P-SYM2-CPT	PhysicalAction
SYM4-L-012	F-WEAK-NEUTRAL-CURRENT-TREE-3GEN	weak neutral current	exact	quotient-nontrivial physical action when defined; target/sector indexed	tree-level neutral current; FCNC=False; offdiag=(2.2e-16,4.05e-17)	P-SYM2-CPT	PhysicalAction
SYM4-L-016	F-CKM-GENERIC-3GEN	quark charged current	violated	quotient-nontrivial physical action when defined; target/sector indexed	J=2.9295157e-05; det[Hu,Hd]=237448.08	Im P-SYM2-CPT	PhysicalAction; DerivedWitness
SYM4-L-020	F-CKM-CPZERO-3GEN	quark charged current	exact	quotient-nontrivial physical action when defined; target/sector indexed	J=-0; Im det[Hu,Hd]=0	P-SYM2-CPT	PhysicalAction
SYM4-L-024	F-CKM-2GEN	quark charged current	exact	quotient-nontrivial physical action when defined; target/sector indexed	J=0; Im det[Hu,Hd]=0	P-SYM2-CPT	PhysicalAction
SYM4-L-028	F-CKM-DEGENERATE-UC	quark charged current	exact	quotient-nontrivial physical action when defined; target/sector indexed	J=0; Im det[Hu,Hd]=0	P-SYM2-CPT	PhysicalAction
SYM4-L-032	F-DEGENERATE-2D-AUTOMORPHISM	degenerate flavor control	exact	quotient-nontrivial physical action when defined; target/sector indexed	fixed CPT premise and same-target CP/ T relation	P-SYM2-CPT	PhysicalAction
SYM4-L-036	F-STRONGCP-GENERIC	QCD theta	violated	quotient-nontrivial physical action when defined; target/sector indexed	bar-theta=0.8500000000000125; periodicity=6.283185307179586; uum=invariant	pe-vac P-SYM2-CPT	PhysicalAction; DerivedWitness
SYM4-L-040	F-STRONGCP-THETA0	QCD theta	exact	quotient-nontrivial physical action when defined; target/sector indexed	bar-theta=1.2434497875801753e-14; periodicity=6.283185307179586; uum=invariant	P-SYM2-CPT	PhysicalAction
SYM4-L-044	F-STRONGCP-THETAPI-INVARIANT	QCD theta	exact	quotient-nontrivial physical action when defined; target/sector indexed	bar-theta=-3.1415926535897807; periodicity=6.283185307179586; uum=invariant	pe-vac P-SYM2-CPT	PhysicalAction
SYM4-L-048	F-STRONGCP-THETAPI-BROKEN	QCD theta	spontaneously_broken	quotient-nontrivial physical action when defined; target/sector indexed	bar-theta=-3.1415926535897807; periodicity=6.283185307179586; uum=spontaneously_broken	pe-vac P-SYM2-CPT	PhysicalAction; DerivedWitness
SYM4-L-052	F-STRONGCP-MASSLESS-U	QCD theta	exact	quotient-nontrivial physical action when defined; target/sector indexed	bar-theta=None; periodicity=6.283185307179586; uum=invariant	periodic-vac P-SYM2-CPT	PhysicalAction
SYM4-L-056	F-HIGGS-ONE-DOUBLET	one-Higgs-doublet scalar	exact	quotient-nontrivial physical action when defined; target/sector indexed	real one-doublet potential; gauge-invariant vacuum orbit	P-SYM2-CPT	PhysicalAction
SYM4-L-060	F-NU-MASSLESS	leptonic charged current	exact	quotient-nontrivial physical action when defined; target/sector indexed	Josc=0; Majorana invariants=[]	P-SYM2-CPT	PhysicalAction
SYM4-L-064	F-NU-DIRAC	leptonic charged current	violated	quotient-nontrivial physical action when defined; target/sector indexed	Josc=0.029977694; Majorana invariants=[]	P-SYM2-CPT	PhysicalAction; DerivedWitness
SYM4-L-068	F-NU-MAJORANA-WEINBERG	leptonic charged current	violated	quotient-nontrivial physical action when defined; target/sector indexed	Josc=0.024129829; Majorana invariants=[-0.07955232138350578, 0.01218800169533607]	P-SYM2-CPT	PhysicalAction; DerivedWitness
SYM4-L-070	F-BVV-TRIPLE-FAKE	B to VV decay	exact	quotient-nontrivial physical action when defined; target/sector indexed	raw=0.2898980; fake=0.2898980	true=0.0000000; P-SYM2-CPT	PhysicalAction
SYM4-L-073	F-BVV-TRIPLE-TRUE	B to VV decay	violated	quotient-nontrivial physical action when defined; target/sector indexed	raw=0.3228102; fake=0.0535344	true=0.2692759; P-SYM2-CPT	PhysicalAction; DerivedWitness
SYM4-L-079	A-SYM4-INTEGRATED-LEDGER	integrated ledger	sector-indexed	quotient-nontrivial physical action when defined; target/sector indexed	target-relative transformation and sector certificate	P-SYM2-CPT	PhysicalAction

19.4 Direct and indirect certification

A direct transition test has stronger action-level standing than an isolated kinematic correlation. A CPT-linked inference can be exact in the declared target but depends on the imported premise packet. Neither route supersedes the other; they answer different evidentiary questions.

T ledger endpoint

T status is antiunitary, sector- and channel-indexed, with explicit ancestry. The ledger can say whether a claim is direct, CPT-linked, diagnostic-only or incomplete.

CP Ledger

20.1 Reader-facing result

CP is exact in the fixed electromagnetic projection and in the electroweak gauge kinetic and one-doublet scalar projections. It is controlled by $\bar{\theta}$ in QCD, by complete weak-basis invariants in the quark Yukawa sector, by the declared neutrino target and PMNS quotient in the lepton sector, and by the gauge-invariant selected vacuum in scalar sectors. Extended Higgs and higher-operator claims are target updates.

20.2 Where CP violation resides

The core Standard Model has two logically independent CP-odd parameter loci:

1. the quark Yukawa/mixing mismatch, represented invariantly by the CKM/Jarlskog structure;
2. the strong invariant $\bar{\theta}$.

A neutrino mass extension can add a Dirac oscillation phase and, in a Majorana target, additional phase loci. The one-doublet scalar potential does not add a spontaneous-CP phase. These statements concern structure and status, not the numerical values of the parameters.

20.3 Canonical CP rows

Table 20.1: CP sector ledger.

Record	Target	Sector/channel	Physical status	Quotient	Invariant certificate	CPT dependency	Role
SYM4-L-002	F-QED-VECTORLIKE-REALMASS	QED	exact	quotient-nontrivial physical action when defined; target/sector indexed	fixed CPT premise and same-target CP/T relation	P-SYM2-CPT	PhysicalAction
SYM4-L-006	F-WEAK-CHIRAL-1GEN	weak charged current	exact	quotient-nontrivial physical action when defined; target/sector indexed	fixed CPT premise and same-target CP/T relation	P-SYM2-CPT	PhysicalAction
SYM4-L-010	F-WEAK-NEUTRAL-CURRENT-TREE-3GEN	weak neutral current	exact	quotient-nontrivial physical action when defined; target/sector indexed	tree-level neutral current; FCNC=False; offdiag=(2.2e-16,4.05e-17)	P-SYM2-CPT	PhysicalAction
SYM4-L-014	F-CKM-GENERIC-3GEN	quark charged current	violated	quotient-nontrivial physical action when defined; target/sector indexed	J=2.9295157e-05; det[Hu,Hd]=237448.08	Im P-SYM2-CPT	PhysicalAction; DerivedWitness
SYM4-L-018	F-CKM-CPZERO-3GEN	quark charged current	exact	quotient-nontrivial physical action when defined; target/sector indexed	J=-0; Im det[Hu,Hd]=0	P-SYM2-CPT	PhysicalAction
SYM4-L-022	F-CKM-2GEN	quark charged current	exact	quotient-nontrivial physical action when defined; target/sector indexed	J=0; Im det[Hu,Hd]=0	P-SYM2-CPT	PhysicalAction
SYM4-L-026	F-CKM-DEGENERATE-UC	quark charged current	exact	quotient-nontrivial physical action when defined; target/sector indexed	J=0; Im det[Hu,Hd]=0	P-SYM2-CPT	PhysicalAction
SYM4-L-030	F-DEGENERATE-2D-AUTOMORPHISM	degenerate flavor control	exact	quotient-nontrivial physical action when defined; target/sector indexed	fixed CPT premise and same-target CP/T relation	P-SYM2-CPT	PhysicalAction
SYM4-L-034	F-STRONGCP-GENERIC	QCD theta	violated	quotient-nontrivial physical action when defined; target/sector indexed	bar-theta=0.8500000000000125; periodicity=6.283185307179586; vacuum=invariant	P-SYM2-CPT	PhysicalAction; DerivedWitness
SYM4-L-038	F-STRONGCP-THETA0	QCD theta	exact	quotient-nontrivial physical action when defined; target/sector indexed	bar-theta=1.2434497875801753e-14; periodicity=6.283185307179586; vacuum=invariant	P-SYM2-CPT	PhysicalAction
SYM4-L-042	F-STRONGCP-THETAPI-INVARIANT	QCD theta	exact	quotient-nontrivial physical action when defined; target/sector indexed	bar-theta=-3.1415926535897807; periodicity=6.283185307179586; vacuum=invariant	P-SYM2-CPT	PhysicalAction
SYM4-L-046	F-STRONGCP-THETAPI-BROKEN	QCD theta	spontaneously_broken	quotient-nontrivial physical action when defined; target/sector indexed	bar-theta=-3.1415926535897807; periodicity=6.283185307179586; vacuum=spontaneously_broken	P-SYM2-CPT	PhysicalAction; DerivedWitness
SYM4-L-050	F-STRONGCP-MASSLESS-U	QCD theta	exact	quotient-nontrivial physical action when defined; target/sector indexed	bar-theta=None; periodicity=6.283185307179586; vacuum=invariant	P-SYM2-CPT	PhysicalAction
SYM4-L-054	F-HIGGS-ONE-DOUBLET	one-Higgs-doublet scalar	exact	quotient-nontrivial physical action when defined; target/sector indexed	real one-doublet potential; gauge-invariant vacuum orbit	P-SYM2-CPT	PhysicalAction
SYM4-L-058	F-NU-MASSLESS	leptonic charged current	exact	quotient-nontrivial physical action when defined; target/sector indexed	Josc=0; Majorana invariants=[]	P-SYM2-CPT	PhysicalAction
SYM4-L-062	F-NU-DIRAC	leptonic charged current	violated	quotient-nontrivial physical action when defined; target/sector indexed	Josc=0.029977694; Majorana invariants=[]	P-SYM2-CPT	PhysicalAction; DerivedWitness
SYM4-L-066	F-NU-MAJORANA-WEINBERG	leptonic charged current	violated	quotient-nontrivial physical action when defined; target/sector indexed	Josc=0.024129829; Majorana invariants=[-0.07955232138350578, 0.01218800169533607]	P-SYM2-CPT	PhysicalAction; DerivedWitness
SYM4-L-069	F-BVV-TRIPLE-FAKE	B to VV decay	exact	quotient-nontrivial physical action when defined; target/sector indexed	raw=0.2898980; true=0.0000000; fake=0.2898980	P-SYM2-CPT	PhysicalAction
SYM4-L-072	F-BVV-TRIPLE-TRUE	B to VV decay	violated	quotient-nontrivial physical action when defined; target/sector indexed	raw=0.3228102; true=0.2692759; fake=0.0535344	P-SYM2-CPT	PhysicalAction; DerivedWitness
SYM4-L-076	A-SYM4-INTEGRATED-LEDGER	integrated ledger	sector-indexed	quotient-nontrivial physical action when defined; target/sector indexed	target-relative transformation and sector certificate	P-SYM2-CPT	PhysicalAction

20.4 Generalized CP and vacuum status

A CP-conserving target may have complex representatives if a generalized CP map exists. A coupling-level CP symmetry may coexist with a CP-violating selected vacuum in an extended scalar target. The core one-doublet target does not realize that second possibility. The action, couplings, vacuum and invariant diagnostics remain separate rows.

20.5 Core question answered

The atlas can answer:

- which sectors violate CP by architecture, parameter or vacuum;
- which parameter combination is physical;
- which basis changes are skin;
- which invariant certificate is complete for the target;
- whether T status is direct or CPT-linked;
- whether a claim is core, parameter-limit, frontier or target update.

CP ledger endpoint

There is no single CP bit for the Standard Model. The canonical answer is a synchronized set of sector rows in which quark mixing, strong $\bar{\theta}$, neutrino variants, scalar vacuum and anomaly/global-form data have separate causal ancestry and target status.

Cross-Sector Compiler and Projection–Recomposition

21.1 From sector rows to a theory-level answer

A list of correct local statements does not by itself constitute an integrated ledger. The compiler must verify that rows sharing a field, parameter or theorem interface use the same target data. The quark charged-current CP row and the neutral-current flavor row must use the same Yukawa quotient. The strong-CP row and axial-anomaly row must use the same mass convention and topological target. The scalar-vacuum row must use the same Higgs carrier and gauge quotient as the Yukawa mass-response row. The CP–T relation must use the same CPT packet across all sectors.

21.2 Compiler invariants

Recomposition checks the tuple

$$\mathcal{C}_4 = (D, \text{fields}, \text{couplings}, q, \Phi, \Theta, N, V, \mathcal{A}, \text{CPT}, B, R, E).$$

Rows can be combined only if their shared coordinates agree or are connected by an explicit transport. A conflict fails closed.

Cross-sector compiler theorem

The integrated ledger is the unique recomposition of sector rows that preserves all shared target coordinates, presentation quotients, invariant certificates, anomaly imports, vacuum/Higgs interfaces, CPT dependencies and source versions. A row set with a silent target mismatch is not an alternative integrated assignment; it is incompatible.

21.3 No-uniform-bit output grammar

The compiler generates qualified statements of the form:

In target D , operation δ is [exact/violated/spontaneously broken/state dependent/not defined in target] in sector s under parameter and vacuum conditions P , with physical quotient q , certificate I , anomaly status A , CPT dependency C and range R .

It refuses a theory-wide Boolean compression unless every sector row has the same status and the same causal ancestry, which the Standard Model does not.

21.4 Core integrated answer

The integrated result can be summarized without losing its typing:

- C and P are exact in vectorlike electromagnetic/strong projections but violated by chiral weak architecture, including the fixed tree-level neutral-current projection;
- T and CP are exact in those projections only when strong and flavor CP-odd invariants are in exact conserving classes;
- the quark charged-current sector can violate CP/T through the CKM invariant;

- QCD can violate P/T/CP through $\bar{\theta}$ while preserving C;
- the real one-doublet scalar potential and selected vacuum are C/P/T/CP compatible at the isolated scalar locus and introduce no independent spontaneous-CP phase;
- leptonic CP depends on the neutrino target;
- CPT remains an exact theorem/action only under the fixed SYM-2 premise packet;
- global-form and generalized-operator questions remain explicit frontiers.

21.5 Compiled ledger data product

The complete machine-readable ledger contains 79 atomic rows. The full table appears in the Technical Audit Companion; the main text presents the operation-specific projections above. Every row has a stable identifier, target, sector, quotient, invariant certificate, CPT dependency, role, claim status and nonclaim.

Exact falsification route

A proposed new sector row reopens the compiler if it preserves the relevant closed target and the assembly transport rules and changes the theory-level qualified answer without being equivalent to an existing row, a presentation-only refinement, a diagnostic already represented, or a declared target update.

Part IV

Finite Normal-Form Closure

Executed Target-Denial Burdens and the Governance Quotient

22.1 Baseline query vectors

The role architecture is not a thematic outline and is not supplied as a list of names. Each Reference-complete target carries an executable query vector. Depending on the target, queries test domain fixation, action definition, projective implementation, sector status, CPT compatibility, basis quotient, mass diagonalization, invariant completeness, θ -mass synchronization, vacuum realization, neutrino phase locus, triple-product ancestry, or assembly routing.

For each opaque target slot s , the engine constructs D_{λ}^{-s} and reruns every query. The denial record includes the before and after result, the exact set of lost or changed queries, and whether the denial creates an incomplete target, a physical failure, or an explicit target update. This calculation is stored in the denial-transition ledger and can be replayed without loading any candidate registry.

22.2 Governance quotient

A raw burden is determined by its loss tuple

$$\mathbf{b}_s = (L_s, \Delta_s, C_s, F_s, U_s).$$

Two burdens are governance-equivalent only when those tuples agree. The quotient

$$R_{\lambda} = B_{\lambda} / \sim_{\text{gov}}$$

removes redundant denials while preserving every distinct theorem-bearing function. The executed family contains 147 target-relative role positions. Their machine-facing records retain opaque slot IDs, lost-query sets, denial profiles, and governance-class IDs; human role titles are generated only as an expository projection.

22.3 Multiplicity and exact degeneracy

Lawful multiplicity is not inferred from repeated role names. In the controlled exact-degeneracy target, two target slots have the same denial profile and are physically interchangeable before a basis orientation is selected. The quotient retains the required multiplicity, which later produces multiple compatible assignments and a non-trivial automorphism group. Duplicating an unrelated burden with no additional denial loss produces no new governance class and is rejected as redundant.

SYM4-T28 – Extensional occupant-independent role necessity

Relative to each fixed target and its executable query grammar, every canonical role is forced by a recorded loss of theorem-bearing consequence. Role positions and lawful multiplicities are fixed before candidate inspection. A complete same-target denial behavior absent from the governance quotient is a role-completeness counterexample and reopens closure.

Claim status. Locally proved by exhaustive denial evaluation over every registered target slot.

Exact falsification route

A target coordinate whose removal changes a theorem query but produces no burden, or a burden that can only be described by naming its eventual occupant, invalidates the role-completeness claim.

Role-Blind Constructor Fixed Point

23.1 Primitive physical seeds

The physical universe begins with imported or target-primitive QFT objects: state and field transformation laws, Yukawa matrices, θ and mass data, Higgs and vacuum data, neutrino mass structures, transversity amplitudes, and the integrated target registry. A seed contains no role, eligibility, preferred assignment, or compatibility verdict.

23.2 Formula-bearing constructor registry

The authoritative registry contains 45 constructors. Representative families include:

- domain, action, ray, sector, and CPT packet construction;
- quark mass diagonalization, CKM mismatch, weak-basis equivalence, and CP-invariant computation;
- exact-degeneracy projectors and stabilizer axes;
- $\bar{\theta}$ construction, axial transport, special- θ vacuum packets, and negative controls;
- Higgs carrier, gauge-equivalence, vacuum-orbit, and CP-status packets;
- massless, Dirac, and Majorana neutrino mixing, equivalence, and invariant packets;
- transversity true/fake triple products, inverse-transition packets, and deliberately false diagnostic controls;
- target-indexed ledger compilation.

Each constructor has typed inputs and outputs, an applicability predicate, an executable formula, a proposition path, inherited limitations, and failure behavior.

23.3 Fixed-point calculation

Starting from the primitive seeds, the engine repeatedly applies every applicable constructor. Each output carries its constructor ID, parents, depth, formula result, source propositions, and content digest. The calculation terminates after maximum depth 5 with 213 raw artifacts. Negative-control constructors remain in the universe as same-target rivals; they are not deleted merely because they fail later compatibility.

The constructor audit verifies that every nonseed artifact has a registered derivation, every registry entry is actually exercised where applicable, no unsupported final occupant is inserted manually, and grammar mutation changes the fixed point or routes to an explicit failure.

SYM4-T29 – Registered constructor saturation and role-blind physical generation

For every closed target, the registered physical-constructor grammar reaches a finite fixed point. Every non-primitive artifact has an executed derivation path and inherited source limitations. Candidate formation is independent of the burden architecture, and no final role occupation is supplied as a primitive object.

Exact falsification route

A physically applicable same-target constructor absent from the registry, an artifact with no derivation path, or a constructor whose output depends on a burden or desired assignment reopens candidate completeness.

Executable Presentation-Witness Groupoid

24.1 Map families

Presentation equivalence is generated by executable group actions, not by similar display names and not by a finite catalogue of sample phases. The certificate registers target-relative parameterized groups for quark rephasings, Dirac-lepton rephasings, the restricted Majorana presentation group, anomaly-aware axial rotations, Higgs gauge phases, projective ray phases, and the exact-degeneracy stabilizer. Finite arrow records serve as validation witnesses; orbit membership is decided by the corresponding solver or canonicalizer for arbitrary admitted parameters.

For example, a quark rephasing acts as

$$V \mapsto D_u V D_d^\dagger,$$

and the orbit solver determines whether diagonal phases exist on the support graph of any proposed pair. The Dirac-neutrino target admits a corresponding two-sided lepton rephasing, whereas the Majorana target permits arbitrary charged-lepton phases but only the target-licensed discrete signs on the neutrino side. An axial solver verifies the complete simultaneous transformation of the quark mass phases, θ , and anomaly data while preserving $\bar{\theta}$. A Higgs solver canonicalizes every nonzero one-doublet vacuum representative to the same gauge orbit. At exact up-charm degeneracy the solver composes arbitrary $U(2)_{uc}$ stabilizer transformations with down-quark rephasings.

24.2 Arrow validation

Each finite witness arrow contains its target, domain and codomain artifact types, forward parameters, inverse parameters, and source propositions. In addition, every continuous family has a parameter-independent orbit-membership or canonicalization routine. The audit executes both the finite groupoid axioms and large deterministic sets of previously unregistered group parameters:

1. the forward map;
2. the inverse map;
3. inverse-after-forward and forward-after-inverse closure;
4. composition through actual functions;
5. preservation of the complete physical invariant signature;
6. descent of eligibility and compatibility to the resulting class.

Corrupting an inverse or deleting an invariant-preservation condition must fail the corresponding audit test.

24.3 Narrow quotient discipline

The quotient removes only presentation multiplicity. Target identity is computed after canonicalization of the witnessed orbit, while a separate representative digest records that a nonidentity map has changed the display. Exact-degeneracy transformations are therefore absorbed only through the admitted stabilizer action; distinct nondegenerate mixing angles remain different physical points. The quotient does not identify:

- a cyclicly permuted CKM control that changes the immutable physical-point fingerprint and is therefore excluded before presentation quotienting;

- $\bar{\theta} = 0$ with $\bar{\theta} = \pi$;
- an action-invariant π vacuum with a spontaneously CP-broken π vacuum;
- Dirac with Majorana neutrino targets;
- a fake absorptive triple product with a true weak-phase triple product;
- a one-doublet Higgs target with an extended scalar sector.

SYM4-T30 – Witnessed presentation groupoid

The registered finite witnesses and parameterized continuous group actions form a target-indexed presentation groupoid. Orbit membership is extensional over the complete admitted group, rather than restricted to sampled parameters. Its orbits preserve the complete physical tuple. Eligibility, compatibility, and normal-form occupation descend to those orbits, while every difference changing target, invariant, vacuum, anomaly, operator locus, or failure ancestry remains distinct.

Exact falsification route

Two artifacts identified despite differing invariant content, or two artifacts kept separate despite a complete invertible full-tuple-preserving map, invalidate the quotient.

Operational Eligibility by Target-Hole Restoration

25.1 No capability matching

The calculus contains no primitive capability vocabulary linking burdens to candidates. For a burden generated by denying slot s , let D_{λ}^{-s} be the denied target and let Q_{lost} be the lost or changed query set. For every same-fiber presentation residue x , the engine attempts a typed insertion:

$$D_{\lambda}^{-s}[x].$$

It then reruns the complete target-query vector.

25.2 Eligibility criterion

A pair is eligible exactly when:

1. the artifact type can inhabit the denied slot without changing target identity;
2. every lost query returns to its baseline status and value;
3. no previously valid query is broken;
4. no undeclared field content, operator category, background, phase, or boundary datum is introduced;
5. the failure and continuation profile agrees with the source target.

Thus

$$E_{\lambda}(b, x) = 1$$

only when x operationally restores the denied theorem behavior. A negative edge records the exact query-status or value mismatch, not a generic semantic rejection.

25.3 Complete all-pairs incidence

All 1311 same-fiber burden-residue pairs are evaluated. 150 pass restoration. The matrix retains every failure witness, including wrong physical action, incomplete CPT packet, incorrect invariant value, unresolved vacuum, wrong neutrino target, and same-target negative controls.

SYM4-T31 – Complete all-pairs operational eligibility

For every closed target and the integrated assembly, every burden-residue pair receives exactly one target-hole-restoration disposition. No positive edge is inferred from an object name, kind, locus label, source-record identity, or shared capability tag.

Exact falsification route

A residue that restores every denied query but receives no edge is a false negative. A residue that fails a baseline query yet receives an edge is a false positive. Either reopens the finite theorem.

Relational Compatibility by Complete Target Reassembly

26.1 Pointwise eligibility is not realization

A set of residues can restore each target hole separately and still fail to coexist as one physical realization. The engine therefore enumerates every complete pointwise-eligible bijection and constructs the reassembled target

$$D_\lambda[\sigma].$$

It reruns every baseline query and every cross-object equation. No primitive candidate-relation graph is supplied, and an unknown relation blocks the assignment rather than defaulting to true.

26.2 Cross-object equations

The relation bundle includes, where applicable:

- mass diagonalization and $V_{\text{CKM}} = U_{uL}^\dagger U_{dL}$;
- agreement between selected mixing data and CP-odd invariants;
- exact-degeneracy stabilizer compatibility;
- $\bar{\theta} = \theta + \arg \det M_q$ and axial-transport invariance;
- action-level special- θ status versus vacuum realization;
- Higgs carrier, gauge orbit, and vacuum CP consistency;
- PMNS reconstruction and Dirac/Majorana rephasing restrictions;
- true/fake triple-product formulas and inverse-transition data;
- CPT premise/action/sector synchronization;
- proposition-hypothesis and target-fingerprint agreement;
- compiler routing across all closed fibers and explicit frontiers.

Each result contains computed left and right values or an exact failure reason.

26.3 Executed rejection

The generic three-generation CKM constructor deliberately generates a cyclicly permuted mixing control. The complete mixing orbit is part of the immutable physical-point fingerprint, so this control changes the closed target and is excluded at target admission rather than carried into same-point compatibility. Complete-target reassembly remains fail-closed for every assignment that survives that admission gate.

Across the complete family, 22 pointwise-eligible assignments are enumerated. 1 fail reassembly and 21 survive.

SYM4-T32 – Relational compatibility

A complete pointwise-eligible assignment realizes the target only when the fully reassembled package reproduces the entire baseline query vector and every applicable cross-object equation. Compatibility is computed

from selected physical artifacts and target coordinates; it is not copied from primitive relation edges.

Exact falsification route

An omitted cross-equation, an assignment accepted with an unresolved relation, or a manually preferred map that bypasses reassembly invalidates compatibility closure.

Lawful Automorphisms and Normal-Form Groupoids

27.1 Certificate graph

The theorem-bearing equivalence is not database identity. For each target, the project constructs a colored relational certificate graph containing target coordinates, burden denial profiles, residue invariant signatures, operational eligibility, reassembly relations, presentation arrows, failure behavior, and continuation data. Display names, source row order, record IDs, and expected occupants are excluded from the invariant signature.

A lawful automorphism is a graph automorphism preserving every theorem-bearing color and relation. Its induced action on compatible assignments is computed rather than declared.

27.2 Degeneracy stress target

The controlled exact-degeneracy target contains two interchangeable axis positions and two physically equivalent axis residues. It has multiple compatible complete assignments. The computed certificate automorphism group contains the identity and the swap action, together with the corresponding incidence actions; the compatible assignments lie in one orbit. This demonstrates that the quotient removes genuine lawful symmetry rather than forcing the automorphism group to be trivial through over-individuating IDs.

27.3 Normal-form groupoid

Let Σ_λ be the set of compatible complete assignments and let $\Gamma_\lambda^{\text{lawful}}$ act by the induced certificate automorphisms. The normal-form groupoid has objects Σ_λ and morphisms generated by that action. The computed endpoint is

$$|\Sigma_\lambda / \Gamma_\lambda^{\text{lawful}}| = 1$$

for every closed target, and one compiler orbit for the separately typed assembly.

SYM4-T33 – Lawful full-invariant automorphism theorem

Every assignment equivalence is induced by an automorphism of the complete certificate graph. Database labels and expected occupants have no authority. The computed group action preserves target, denial structure, physical invariants, eligibility, reassembly compatibility, presentation, continuation, and failure data.

SYM4-T34 – Fiberwise and integrated discrete-symmetry normal form

For every Reference-complete closed target D_λ ,

$$|\pi_0 \mathcal{N}_4(D_\lambda)| = 1.$$

The separately typed integrated ledger assembly has one compiler normal form. It is not counted as an additional closed physical theory.

Exact falsification route

An unaccounted full-certificate automorphism, an overbroad automorphism that erases a physical distinction, or a second connected component reopens the normal-form theorem.

Refinement, Ablation, Projection, and Recomposition

28.1 Destructive-use tests

The audit verifies that proof components perform substantive work rather than merely appearing in files. It executes:

- target-slot denials and query-vector changes;
- constructor omission and applicability mutations;
- inverse-map corruption and overbroad presentation quotient tests;
- restoration-edge deletion and false-edge insertion;
- cross-equation ablation, including CKM reconstruction, $\bar{\theta}$, vacuum, neutrino, CPT, and triple-product relations;
- automorphism-signature ablations;
- source-proposition and target-hypothesis mutations.

A load-bearing deletion must either destroy closure or route the affected theorem to OPEN; a conservative evidence enrichment must preserve the orbit.

28.2 Frozen holdouts

Four maximum-stress cases were frozen after the final constructor grammar hash and before final audit packaging. They cover exact degeneracy, the $\bar{\theta} = \pi$ action/vacuum split, fake absorptive triple products, and the Majorana phase locus. A grammar change invalidates the holdout freeze and requires a newly frozen certificate before the cases can be used again.

28.3 Projection and recomposition

The integrated ledger is a compiler assembly over fixed physical fibers. Each sector projection carries its target ID, proposition path, quotient, invariant, anomaly, vacuum, and CPT dependency. Recomposition succeeds only when those coordinates agree and frontiers remain explicitly routed. It does not turn mutually exclusive neutrino, vacuum, or special- θ variants into one physical theory.

SYM4-T35 – Refinement and projection-recomposition stability

Every registered split, merge, omission, duplication, target update, and conservative-enrichment constructor receives an executed disposition. Witnessed refinements transport to the canonical orbit; a complete nonisomorphic same-target refinement reopens closure rather than being silently absorbed.

SYM4-T36 – Conservative continuation and successor discipline

Evidence-only enrichment that leaves target, query behavior, physical fixed point, presentation quotient, restoration incidence, reassembly compatibility, and lawful automorphisms unchanged preserves the canonical orbit. Standing-changing content produces ClosureReopened or a versioned successor target.

Downstream Interfaces and Successor Discipline

29.1 Why transport rules matter

SYM-4 is a foundation for later sector papers, not a terminal standalone catalogue. Its exports must be stable enough that SYM-5 through SYM-11 cannot silently change a CP, anomaly, vacuum or target field. Every exported record carries an immutable ID, target fingerprint, physical status, role, source packet and exact nonclaims.

Table 29.1: Downstream interfaces and successor discipline.

Successor	Export	Transport rule
SYM-5	basis quotient, generalized CP maps, CKM/PMNS invariant records	Import target and proposition IDs unchanged; any new flavor content creates a successor target.
SYM-6	axial anomaly, chiral/vacuum and strong-phase interfaces	Preserve SYM-3 anomaly ancestry and SYM-4 action/vacuum split.
SYM-7	discrete baryon/lepton remnants	Index by operator dimension, neutrino target, and anomaly status.
SYM-8	one-doublet vacuum CP and Higgs representative quotient	Import Higgs carrier/vacuum split; extended scalars are target updates.
SYM-9	special- θ global form, line/defect sectors and discrete anomalies	Close only after faithful group and operator category are fixed.
SYM-10	scale/order transport and emergent discrete structures	Preserve invariant/representative distinction under RG and matching.
SYM-11	frozen discrete ledger, proof objects, rivals and frontiers	Use the canonical IDs and exact theorem boundary in whole-arc exhaustion.

29.2 SYM-5 interface

SYM-5 receives the weak-basis and rephasing quotient, generalized CP existence conditions, quark and lepton invariant packages, degeneracy stabilizers and the distinction between matrix representation and physical mixing. It may refine the flavor atlas, but any change to the target or invariant completeness must produce a successor record.

29.3 SYM-6 and SYM-8 interfaces

SYM-6 receives the axial anomaly, chiral/vacuum and custodial interfaces. SYM-8 receives the Higgs carrier, one-doublet vacuum result, gauge-fixed representative classification and extended-scalar router. The full Higgs Arc provides a stronger carrier/identity/scale/vacuum/extension boundary than the original ARC plan and is therefore part of the frozen interface.

29.4 SYM-7 interface

Discrete baryon/lepton remnants must import the SYM-3 anomaly status and the SYM-4 discrete-action grammar. A discrete remnant is not exact merely because a continuous phase has been broken; operator dimension, neutrino content and anomaly compatibility remain mandatory.

29.5 SYM-9 interface

Special- θ , faithful global form, line/monopole sectors, higher-group structures, non-invertible defects and discrete anomalies are handed to SYM-9 as explicit frontiers. SYM-9 may close them only by producing successor target

fingerprints and transport maps back to the sharp-local ledger.

29.6 SYM-10 interface

Scale and RG transport must distinguish running description from physical action. A CP-odd operator or phase can be threshold-transported, but numerical values and matching conditions remain target data. Scheme changes are presentation only when physical invariant predictions are preserved.

29.7 SYM-11 interface

SYM-11 receives the frozen discrete ledger, role/candidate groupoid, frontier list and counterexample protocol. Whole-arc exhaustion requires candidate completeness across all sectors, not merely the local singleton theorem proved here.

SYM4-T38 – Downstream transport theorem

A downstream paper importing a SYM-4 record must preserve its target fingerprint, quotient, invariant, CPT, anomaly, vacuum and evidence fields. A lawful refinement creates a versioned successor linked by an explicit transport. Silent modification invalidates the composition chain.

SYM4-T39 – No-second-classifier/apex consequence

After necessary kernel activation, fixed-domain closure, the AMetric boundary package and the local singleton normal form are established, the exact imported apex proposition from *Unus Solus Possibilis Est* excludes a second same-domain governance-non-equivalent classifier. The theorem is recorded as `IMPORTED_THEOREM_APPLIED`, with its premise DAG and source locator explicit. The apex consequence does not choose the local occupant and cannot repair a missing SYM-4 sector or relation.

Part V

Audit, Falsification, and Endpoint

Maximum-Stress Cases and Holdout Calibration

30.1 Purpose of the benchmark suite

The hard-case suite is not a gallery of examples chosen after the classifier is known. It is the minimum adversarial set required to expose the category errors that a one-bit C/P/T/CP catalogue would conceal. Each case changes one or more of the coordinates that the SYM-0 calculus declares logically independent: target, quotient, anomaly, vacuum, topology, operator category, parameter limit, scale, or physical action. The adequacy criterion is therefore stronger than obtaining a plausible verbal answer. One and the same typed grammar must resolve all registered cases without adding a case-specific role, exception value, or untracked target mutation.

Table 30.1: Maximum-stress and frozen-holdout suite.

ID	Case	Discrimination	Status	Assertions	Holdout	Executed details
HC01	P in vectorlike QED versus charged- and neutral-current weak interactions	Vectorlike exactness versus chiral violation without a theory-wide bit.	PASS	3	False	QED exact; charged- and neutral-current chiral weak loci violate P
HC02	C and flavor status in vectorlike versus chiral weak sectors	Neutral-current C/P violation is separated from CP/T exactness and from the derived absence of tree-level FCNC.	PASS	9	False	C/P violation is distinct from CP/T exactness and from the derived absence of tree-level FCNC
HC03	Complex Yukawa matrices with all physical CP invariants zero	Complex coefficients versus complete invariant certificate.	PASS	5	False	complex_presentation=True; J=-0.0
HC04	Generic three-generation CKM CP violation	Nonzero Jarlskog and commutator witness after quotient.	PASS	9	False	J=2.9295157e-05; cyclic contestant pointwise admitted and rejected by complete Yukawa/mixing reassembly
HC05	Two-generation quarks	No physical Dirac-type CP phase.	PASS	3	False	phase_count=0 and J=0
HC06	Exact mass degeneracy	Stabilizer enlargement and phase collapse.	PASS	14	True	unseen U(2) angle=0.719, phase=0.337; complete composed witness=True; merged=True
HC07	Axial redefinition transporting theta and mass phases	Presentation transport preserving bar-theta.	PASS	2	False	validated axial arrows=10
HC08	Exactly massless quark target	Bar-theta parameter removability separated from exact C/P/T/CP actions.	PASS	6	False	bar-theta parameter removable; C/P/T/CP physical actions exact
HC09	bar-theta zero	Action-level P/T/CP invariant strong target.	PASS	4	False	bar-theta=0; P/T/CP exact
HC10	bar-theta pi	Action fixed point separated from invariant or spontaneously broken vacuum branch.	PASS	12	True	unseen mass phases=[0.083, -0.219, 0.391]; independently derived action fixed point with invariant and CP-exchanged vacuum branches
HC11	One-Higgs-doublet vacuum	No physical independent spontaneous-CP phase in the core target.	PASS	6	False	gauge representative phase is quotient-trivial
HC12	Extended-Higgs spontaneous CP	Explicit target update, not core repair.	PASS	2	False	routed to explicit successor target
HC13	Antiunitary T versus naive reversal	Ray action, inverse transition, and kinematic sign reversal remain distinct.	PASS	4	False	ray action, inverse transition and kinematic map are distinct
HC14	Absorptive fake T-odd correlation	Nonzero raw triple product with zero true CP/T combination.	PASS	9	True	baseline raw=0.289898; unseen strong phase=0.527, derived raw=0.183574, true=0, fake=0.183574
HC15	CPT-linked CP/T inference	Exact premise and same-target relation instantiated by a concrete CP-violating target; generic CPT retained as an interface.	PASS	5	False	same-target CPT transport instantiated by a concrete CP-violating quark target; generic CPT object remains a conditional interface
HC16	Minimal massless-neutrino target	No physical PMNS mixing or leptonic CP phase, while CP and T actions remain exact.	PASS	5	False	no physical PMNS or leptonic phase; CP/T actions remain exact
HC17	Dirac-neutrino oscillation CP	Rephasing-invariant oscillation quartet.	PASS	4	False	Josc=0.02997769
HC18	Majorana phase locus	Restricted rephasing and LNV-only phase witnesses.	PASS	7	True	baseline phases=[0.6, 1.0]; unseen phases=[0.319, 1.071]; unseen invariants=[-0.1395472333154579, 0.014170211476267095]

continued

ID	Case	Discrimination	Status	Assertions	Holdout	Executed details
HC19	Generalized CP combined with flavor-unitary map	Executable Yukawa invariance equations and unitary generalized-CP matrices.	PASS	8	False	executed $XQ^\dagger Y_u X_u=Y_u^*$ and $XQ^\dagger Y_d X_d=Y_d^*$; residuals=(0,0)
HC20	Discrete anomaly/global-form frontier	SYM-9 successor rather than silent closure.	PASS	2	False	explicit SYM-9 successors
HC21	External background or asymmetric boundary	Target update before P/T comparison.	PASS	2	False	background and boundary must be fixed before P/T comparison
HC22	Dimension-fixed SMEFT CP-odd operator	Operator order and matching-scale target update.	PASS	2	False	operator dimension and matching scale are successor coordinates

30.2 Executed dossier pattern

Every hard-case dossier contains the following proof objects:

1. a target fingerprint, including field content, sector, vacuum, topology, global form, operator locus and parameter regime;
2. an explicit transformation, generalized CP map, transport or diagnostic relation;
3. a physical quotient and a statement of what survives that quotient;
4. a complete or explicitly limited invariant certificate;
5. imported anomaly and CPT interfaces where applicable;
6. a causal-ancestry record distinguishing explicit coupling, anomaly, vacuum, absorptive phase, topology and target update;
7. a role-blind physical candidate packet;
8. positive and negative role tests;
9. exact nonclaims and reopening conditions.

Missing information routes to `IncompleteCertificate`, `OpenBurden`, or `TargetUpdate`; it never defaults to “conserved”, “violated”, “skin”, or “physical action”.

30.3 Representative resolved cases

30.3.1 Complex Yukawa representatives with vanishing physical invariants

A weak basis may display complex entries even when the target admits a generalized CP map and the complete target-relative CP-odd invariant set vanishes. The complex entries are presentation facts; the existence of the compatible antiunitary map and the invariant certificate determine the physical disposition. The case defeats the inference

$$\text{complex coefficient} \implies \text{physical CP violation.}$$

It is resolved jointly by the weak-basis quotient, generalized-CP criterion and invariant-completeness relation [Kobayashi and Maskawa, 1973, Jarlskog, 1985, Maley, 2026t,u].

30.3.2 Exact degeneracy

An exact mass degeneracy enlarges the stabilizer of the mass operator. Rotations within the degenerate subspace can remove parameters that would be physical in the generic nondegenerate target. The case therefore tests whether the candidate generator knows the target stabilizer before counting phases or angles. It also supplies the controlled nontrivial automorphism target used in the finite proof.

30.3.3 Axial redefinition and strong CP

An axial redefinition does not act on the mass phase while leaving the path-integral measure and theta term untouched. The case is resolved only when the anomaly transport, mass determinant and topological term are synchronized. The invariant object in the declared convention is $\bar{\theta} = \theta + \arg \det M_q$, not either summand separately [Adler, 1969, Bell and Jackiw, 1969, Fujikawa, 1979, Maley, 2026n,o].

30.3.4 T-odd observables with absorptive phases

A triple-product correlation or other naively T-odd quantity can be nonzero because of final-state or absorptive phases even when no fundamental T-violating weak phase is present. The case forces separation of the physical antiunitary T operation, a process-reversed transition, a kinematic sign reversal and an observable diagnostic. A direct T test compares appropriately exchanged initial and final states; it is not licensed merely by the sign of one correlation [CPLEAR Collaboration, 1998, BaBar Collaboration, 2012].

30.3.5 One-doublet vacuum and extended-Higgs update

The one-doublet core contains no independent gauge-invariant spontaneous-CP phase. A second doublet or another scalar sector can create additional invariant relative phases and vacuum structures, but that is an explicit target update. The paired cases verify that the classifier neither imports extended-Higgs possibilities into the core nor forbids them as enlarged-domain physics [Lee, 1973, Maley, 2026k,l,m].

30.3.6 Dirac and Majorana neutrino targets

Ordinary oscillation probabilities depend on Dirac-type rephasing invariants and are insensitive to Majorana phases. A Majorana target has a more restrictive field-rephasing quotient, so additional phase loci can be physical in lepton-number-violating observables. Treating the two targets as one record either overcounts oscillation information or erases genuine Majorana structure [Maki et al., 1962, Schechter and Valle, 1980, Bilenky et al., 1980, Maley, 2026p,q].

30.4 Frontier cases are exact dispositions

The special-theta/global-form case and the discrete-anomaly case are marked `frontier_explicit`, not unresolved by accident. Their exact disposition states which missing data prevent local closure and which successor module owns them. This is a stronger result than an unqualified tentative answer: the target boundary and reopening rule are part of the theorem object.

30.5 Holdout test

One registered hard case was withheld from the role and constructor design pass and admitted only after the baseline grammar, quotient and compatibility relations had been frozen. The holdout tests whether the classifier generalizes to a new combination of already-declared physical coordinates rather than merely memorizing its calibration set. The executable audit records one holdout test and one pass. The result is evidence for grammar adequacy at the declared scope; it is not an open-world candidate-exhaustion theorem.

Hard-case adequacy theorem

All 22 registered maximum-stress cases receive a schema-complete disposition under the frozen target, kind, quotient, invariant, anomaly, vacuum, topology, scale, CPT and role grammar. No case introduces a new status value or a one-off role. The two frontier cases route to explicit successor burdens rather than being silently counted as closed.

Exact falsification route

The adequacy result is reopened by a same-target hard case that requires a new physical coordinate, cannot be represented without merging non-identical loci, or receives two incompatible dispositions from the existing rules. A changed field content, global form, operator category, vacuum or scale is instead evaluated through the target-update protocol.

Adversarial Validation and Falsification Standard

31.1 Fixed acceptance standard

The theorem is evaluated under a fixed audit standard. A blocking objection must identify a false physical proposition, hidden target change, missing same-target object, defective presentation quotient, false restoration or compatibility result, incorrect automorphism calculation, incomplete evidence path, reproducibility failure, or overstatement of theorem strength. Preferences for different terminology, broader neighboring targets, numerical predictions, or additional successor sectors do not alter the declared endpoint.

31.2 Executable adversarial classes

The certificate contains permanent negative controls for each theorem-bearing layer:

1. target-coordinate and complete-packet substitutions must change the content-addressed physical-point identity or fail target consistency;
2. CP and T statuses must satisfy the same-target CPT relation whenever the exact CPT premise packet is admitted;
3. action-level symmetry and vacuum realization are computed independently, including the two $\bar{\theta} = \pi$ vacuum branches;
4. target kinds are checked against the actual quark and neutrino mass operators rather than accepted from labels;
5. physically distinct zero-phase mixing matrices remain distinct unless an explicit presentation arrow relates them;
6. every lawful presentation arrow preserves the full physical target fingerprint while its representative hash may change;
7. generalized CP is tested through the explicit Yukawa equations $X_Q^\dagger Y_u X_u = Y_u^*$ and $X_Q^\dagger Y_d X_d = Y_d^*$;
8. every complete assignment is reassembled and subjected to the full target query vector and cross-object equations;
9. imported apex consequences are licensed only by target-specific premise packets and a completed local singleton normal form;
10. the distributed research object must reproduce byte-for-byte from its first clean build and expose one immutable manuscript, companion, and archival-reference tuple.

31.3 Substantive objections to the physics ledger

31.3.1 "CPT makes a separate T ledger unnecessary"

CPT supplies a relation under a premise packet; it does not identify CP and T operations, direct inverse-transition tests, or T-odd diagnostics. Their targets and quotients remain distinct [Maley, 2026r, Greaves and Thomas, 2014, CPLEAR Collaboration, 1998, BaBar Collaboration, 2012].

31.3.2 “Every T-odd correlation demonstrates T violation”

The transversity target contains a nonzero raw T-odd correlation generated by absorptive phases while the true weak-phase combination vanishes. This supplies an explicit counterexample to the inference [Datta and London, 2004].

31.3.3 “A complex Yukawa entry is already physical CP violation”

Weak-basis and rephasing transformations change complex entries while preserving the complete invariant packet. Physical CP standing follows from target-relative generalized-CP or invariant conditions, not display complexity [Jarlskog, 1985, Grimus and Rebelo, 1997].

31.3.4 “The one-doublet Higgs phase proves spontaneous CP violation”

A gauge-fixed Higgs phase is presentation. The one-doublet target is classified through its gauge-invariant carrier and vacuum orbit. Additional scalar structure is a successor target rather than an unregistered same-scope phase.

31.3.5 “Three generations should be derived here”

Generation count is fixed-boundary envelope data. Changing it changes field-role multiplicities, flavor slots, Yukawa dimensions, and mixing structure; SYM-4 does not convert a parameter-count theorem into a numerical generation prediction.

Falsification boundary

A same-target counterexample must preserve the complete content-addressed target point while exposing a missing applicable constructor, false presentation map, missing restoration edge, omitted reassembly equation, incorrect lawful automorphism, or second normal-form component. A changed scalar sector, neutrino mass type, global form, operator order, boundary condition, or numerical benchmark point is a successor target unless already parameterized by the fixed target family.

Audit Architecture, Evidence, and Reproducibility

32.1 Controlled proof objects

The controlled archive contains immutable or generated registries for targets, theorem queries, denial transitions, governance classes, physical seeds, constructor definitions, constructor derivations, presentation arrows, presentation residues, operational eligibility, pointwise assignments, reassembly failures, compatible assignments, automorphisms, orbits, source propositions, theorem evidence, hard cases, nonclaims, and successor interfaces. Manuscript tables are generated projections of those canonical CSV/JSON objects.

32.2 Proposition-level source discipline

Each imported proposition records an exact statement, hypotheses, conclusion, source title, version, locator, target specialization, limitation, source-byte status, and downstream consumers. Imported QFT, imported AASC, and locally executed finite conclusions remain different claim classes. In particular:

- T19 is supported by the concrete triple-product analysis;
- T20 imports the exact SYM-2 CPT interface;
- T22 imports the Higgs carrier and vacuum propositions;
- T25 imports a Majorana-specific invariant proposition.

No paper-level alias is accepted as the terminal evidence path.

32.3 Independent canonical-endpoint replay

A code-separated reference-normalized implementation reads the frozen point-target packets and independently replays the canonical target-level role and residue counts, all-pairs incidence counts, complete assignment counts, compatibility dispositions, lawful automorphism counts, and assignment-orbit counts. It does not import the primary role, candidate, residue, eligibility, assignment, or orbit registries before completing that replay. It does not independently reconstruct the primary raw-constructor fixed point, derivation DAG, presentation-variant universe, or witnessed raw-to-canonical quotient; those stages remain certified by the primary derivation-path, holdout, executable-map, inverse, composition, and overbroad-quotient audits.

SYM4-T37 – Proposition-level source-path completeness and independent canonical-endpoint count replay

Every theorem-bearing finite conclusion has a path from a fixed target and target-matched propositions through denial or constructor formation, presentation quotient, operational eligibility, reassembly compatibility, and lawful orbit calculation. A separately coded implementation independently replays the canonical target-level role and residue counts, all-pairs incidence counts, complete assignment counts, compatibility dispositions, lawful automorphism counts, and orbit counts from the frozen point-target packets. It does not independently reconstruct the primary raw-constructor fixed point, derivation DAG, presentation-variant universe, or witnessed raw-to-canonical quotient; those stages are certified by the primary executable audit, mutation tests, holdouts, and groupoid checks. Imported continuum theorems remain explicit premises rather than software-generated facts.

32.4 Executable audit groups

The final suite contains 4762 tests under the fixed falsification standard. It checks baseline integrity, target and scope fidelity, QFT correctness, quotient fidelity, evidence integrity, semantic and process independence, constructor saturation, finite proof correctness, hard-case adequacy, refinement and ablation behavior, claim-strength generation, and exact reproducibility.

The destructive tests mutate target coordinates, constructor applicability, map inverses, restoration results, reassembly equations, automorphism signatures, source hypotheses, and theorem dependencies. A mutation is useful only when the affected conclusion changes or fails closed.

Table 32.1: Frozen-standard executable audit summary.

Audit group	Tests	Passed	Failed
BASELINE_INTEGRITY	229	229	0
PRIMARY_ADVERSARIAL_REGRESSION	8	8	0
EXTENDED_ADVERSARIAL_REGRESSION	9	9	0
CLAIM_STRENGTH	57	57	0
CONSTRUCTOR_SATURATION	328	328	0
EVIDENCE_INTEGRITY	159	159	0
FINITE_PROOF	1506	1506	0
HARD_CASES	23	23	0
INDEPENDENCE	378	378	0
PRESENTATION_MODULE_INTEGRATION	84	84	0
QFT_CORRECTNESS	46	46	0
QUOTIENT_FIDELITY	1489	1489	0
REFINEMENT_ABLATION	147	147	0
REPRODUCIBILITY	11	11	0
TARGET_SCOPE	277	277	0
V4_4_ROUND1_RESPONSE	11	11	0

32.5 Generated theorem status

The capstone and all local closure theorems are assigned status by dependency expressions. For example, T34 requires passing role generation, constructor fixed point, executable presentation, complete operational eligibility, complete reassembly, automorphism computation, and singleton orbit. T40 additionally requires complete evidence, the independent canonical-endpoint replay, hard cases, frozen-review replay, and deterministic packaging. A failed dependency changes the theorem status to `OPEN`; no prose override exists.

32.6 Deterministic archival build

The build fixes locale, timezone, source epoch, PDF creation and modification dates, trailer identifiers, ZIP timestamps, permissions, entry order, and compression. Two clean builds in independent directories must produce identical SHA-256 values for the scientific manuscript, Technical Audit Companion, combined archival reference, and project ZIP. The archival manifest is generated only from final artifacts and excludes self-referential hash cycles.

32.7 Formalization boundary

The formal target is the finite classification architecture. The project does not formalize interacting path integrals, prove the CPT theorem in a proof assistant, or solve nonperturbative QCD. It machine-checks target fingerprints, query behavior, constructor derivations, presentation maps, all-pairs restoration, full reassembly, automorphisms, orbit components, evidence paths, and exported ledgers.

Audit result

The final certificate reports 4762/4762 tests passed, 22/22 hard cases passed, all frozen holdouts passed, complete source-integrity checks, presentation-covariant target fingerprints, a self-contained first clean build, and byte-identical archival artifacts.

Conclusion: Sector-Indexed Discrete-Symmetry Normal-Form Closure

The Standard Model does not possess one faithful theory-wide C, P, T, or CP bit. A discrete operation can be exact in one sector, violated in another, physical on states, representative on bases, anomalous in a background coupling, dependent on a vacuum, or meaningful only in a parameter-limit target. SYM-4 makes those qualifications compositional rather than ad hoc.

The public QFT ledger separates active actions, state and field implementations, physical quotients, sector restrictions, weak-basis and rephasing freedom, generalized CP maps, complete invariant certificates, θ and mass-phase transport, absorptive ancestry, vacuum realization, neutrino convention, anomaly, global form, and operator range. The AASC ledger then records the standing work of those already-established structures. Exactness, physicality, diagnostic power, and primitive role are never inferred from one another.

Kernel necessity is not one of the finite theorem's outputs: every determinate target is already kernel-bearing. The finite theorem is stronger than a descriptive table because it supplies a complete target-relative construction, quotient, incidence, compatibility, and orbit certificate. Occupant-independent roles are generated from executed query losses over opaque target slots. Physical objects are generated independently by 45 formula-bearing constructors. Presentation multiplicity is removed only by executable invertible maps. Every same-fiber pair is tested by target-hole restoration; every complete eligible occupation is tested by full target reassembly; and every lawful automorphism is derived from the full certificate graph. The calculation yields one normal-form component for each of the 19 closed targets.

SYM4-T40 – Publication capstone

Let $D_4^\star$ be the declared indexed SYM-4 target family and let $\mathcal{N}_4(D_\lambda)$ be the groupoid of complete target-role occupations whose objects satisfy operational eligibility and full reassembly compatibility and whose morphisms are full-certificate-preserving lawful automorphisms. For every role-realization-complete closed target,

$$|\pi_0 \mathcal{N}_4(D_\lambda)| = 1.$$

The canonical sector ledger is a presentation of that unique isomorphism class. The theorem is stable under witnessed redescription, conservative evidence enrichment, and registered projection/recomposition. A new same-target physical distinction, applicable constructor, denial burden, reassembly equation, or second non-equivalent orbit reopens closure; changed field content, vacuum, neutrino convention, global form, operator order, or scalar sector creates a successor target.

33.1 What is closed

Within the declared sharp-local renormalizable Standard Model and controlled target family:

- C, P, T, and CP statements have complete sector and channel indices;
- complex representatives are separated from quotient-stable phases;
- generic, lower-generation, CP-zero, and exactly degenerate quark targets have distinct invariant normal forms;
- strong CP is represented by anomaly-synchronized θ /mass data, with both 0 and π treated as action-level fixed points under ordinary 2π periodicity and vacuum realization separately typed;
- direct T action is separated from naive reversal and absorptive T-odd diagnostics;

- explicit and spontaneous CP are separated, with the one-doublet Higgs core closed at its declared scope;
- massless, Dirac, and Majorana neutrino targets are separated;
- anomaly, global-form, and generalized-operator frontiers are explicit;
- every closed finite target has one lawful normal-form component.

33.2 What remains open

SYM-9 must decide faithful global form, line and monopole sectors, higher-group structure, non-invertible defects, and special- θ completions requiring those data. SYM-10 must control scale and EFT transport. Extended Higgs and SMEFT CP sectors require their own fixed targets. SYM-11 must freeze the whole-arc candidate universe and discharge the final same-scope rival ledger.

33.3 Final statement

The strongest defensible endpoint is neither “the Standard Model violates CP” nor “all discrete symmetries are derived.” It is:

Given a Reference-complete Standard Model target, its frozen gauge, spacetime/CPT, anomaly, flavor, Higgs/vacuum, and neutrino interfaces, and the declared query and constructor grammars, every admitted C/P/T/CP claim has one redescription-stable, sector-indexed physical status and structural normal form, unique modulo lawful presentation. No unqualified theory-wide discrete-symmetry bit survives that classification.

That endpoint meets the original ARC theorem boundary, the stronger reference standard established by SYM-0 through SYM-3, and the predeclared non-drifting falsification standard.

Canonical Theorem Inventory

The inventory distinguishes imported QFT propositions, imported AASC consequences, local physical theorems, grammar-relative coverage results, executable finite theorems, transport theorems and explicit frontiers. A theorem's appearance in the same manuscript does not equalize its proof class.

Table A.1: Canonical theorem inventory.

ID	Theorem	Claim class	Generated status
SYM4-T01	Pre-role non-degenerate determinacy	local physical theorem establishing pre-role non-degenerate determinacy	LOCAL_PHYSICAL_THEOREM
SYM4-T02	Necessary kernel activation	imported kernel-necessity theorem with exact local physical instantiation	IMPORTED_THEOREM_INSTANTIATION
SYM4-T03	Reference-complete indexed-target fixation	imported QFT plus local target specialization	IMPORTED_WITH_LOCAL_INSTANTIATION
SYM4-T04	Imported-interface identity and no status laundering	imported QFT plus local target specialization	IMPORTED_WITH_LOCAL_INSTANTIATION
SYM4-T05	Discrete-object and locus noncollapse	imported QFT plus local target specialization	IMPORTED_WITH_LOCAL_INSTANTIATION
SYM4-T06	Sector projection and verified recomposition	imported QFT plus local target specialization	IMPORTED_WITH_LOCAL_INSTANTIATION
SYM4-T07	Active/passive discrete-action separation	imported QFT plus local target specialization	IMPORTED_WITH_LOCAL_INSTANTIATION
SYM4-T08	Antiunitary-T noncollapse	imported QFT plus local target specialization	IMPORTED_WITH_LOCAL_INSTANTIATION
SYM4-T09	CPT relation lattice	imported theorem/consequence with exact local target interface	IMPORTED_WITH_LOCAL_INSTANTIATION
SYM4-T10	Sector-locality and no-uniform-bit theorem	imported QFT plus local target specialization	IMPORTED_WITH_LOCAL_INSTANTIATION
SYM4-T11	Vectorlike/chiral sector theorem	imported QFT plus local target specialization	IMPORTED_WITH_LOCAL_INSTANTIATION
SYM4-T12	Basis and rephasing quotient	imported QFT plus local target specialization	IMPORTED_WITH_LOCAL_INSTANTIATION
SYM4-T13	Generalized-CP existence criterion	imported QFT plus local target specialization	IMPORTED_WITH_LOCAL_INSTANTIATION
SYM4-T14	Complete CP-odd invariant criterion	imported QFT plus local target specialization	IMPORTED_WITH_LOCAL_INSTANTIATION
SYM4-T15	Degeneracy and two-generation phase collapse	imported QFT plus local target specialization	IMPORTED_WITH_LOCAL_INSTANTIATION
SYM4-T16	Strong-CP invariant-combination theorem	imported QFT plus local target specialization	IMPORTED_WITH_LOCAL_INSTANTIATION
SYM4-T17	Axial-redefinition transport theorem	imported QFT plus local target specialization	IMPORTED_WITH_LOCAL_INSTANTIATION

continued

ID	Theorem	Claim class	Generated status
SYM4-T18	Special-theta routing theorem	target-update or transport theorem	IMPORTED_ WITH_LOCAL_ INSTANTIATION
SYM4-T19	T-odd-correlation noncollapse	imported QFT plus local target specialization	IMPORTED_ WITH_LOCAL_ INSTANTIATION
SYM4-T20	CPT-linked T theorem	imported theorem/consequence with exact local target interface	IMPORTED_ WITH_LOCAL_ INSTANTIATION
SYM4-T21	Explicit/spontaneous CP separation	imported QFT plus local target specialization	IMPORTED_ WITH_LOCAL_ INSTANTIATION
SYM4-T22	One-Higgs-doublet scalar/vacuum theorem	imported QFT plus local target specialization	IMPORTED_ WITH_LOCAL_ INSTANTIATION
SYM4-T23	Neutrino target bifurcation	target-update or transport theorem	IMPORTED_ WITH_LOCAL_ INSTANTIATION
SYM4-T24	Dirac oscillation CP theorem	imported QFT plus local target specialization	IMPORTED_ WITH_LOCAL_ INSTANTIATION
SYM4-T25	Majorana phase-locus theorem	imported QFT plus local target specialization	IMPORTED_ WITH_LOCAL_ INSTANTIATION
SYM4-T26	Discrete-anomaly import theorem	imported theorem/consequence with exact local target interface	IMPORTED_ WITH_LOCAL_ INSTANTIATION
SYM4-T27	Global-form and generalized-operator frontier theorem	target-update or transport theorem	IMPORTED_ WITH_LOCAL_ INSTANTIATION
SYM4-T28	Extensional occupant-independent role necessity	local executable finite theorem	PROVED
SYM4-T29	Registered constructor saturation and role-blind generation	local executable finite theorem	PROVED
SYM4-T30	Witnessed presentation groupoid	local executable finite theorem	PROVED
SYM4-T31	Complete all-pairs eligibility	local executable finite theorem	PROVED
SYM4-T32	Relational compatibility	local executable finite theorem	PROVED
SYM4-T33	Lawful full-invariant automorphism theorem	local executable finite theorem	PROVED
SYM4-T34	Integrated and fiberwise discrete-symmetry normal form	local executable finite theorem	PROVED
SYM4-T35	Refinement and projection-recomposition stability	local executable finite theorem	PROVED
SYM4-T36	Conservative certificate continuation and successor discipline	target-update or transport theorem	PROVED
SYM4-T37	Proposition-level source-path completeness and independent canonical-endpoint count replay	local executable finite theorem	PROVED
SYM4-T38	Downstream transport to SYM-5 through SYM-11	target-update or transport theorem	PROVED
SYM4-T39	No-second-classifier/apex consequence	imported theorem/consequence with exact local target interface	IMPORTED_ THEOREM_APPLIED
SYM4-T40	Publication capstone	local executable finite theorem	PROVED

A.1 Composition order

The composition chain is:

target and source propositions $\longrightarrow$ physical/burden construction $\longrightarrow$ presentation quotient
 $\longrightarrow$ eligibility and compatibility $\longrightarrow$ lawful orbit $\longrightarrow$ downstream transport.

The imported apex consequence enters only after the local orbit theorem. It cannot select an occupant or compensate for an omitted physical relation.

Exact Nonclaims

The following boundaries are part of the theorem object and are also stored in the canonical nonclaim registry.

1. No numerical CKM or PMNS angle, Dirac phase, Majorana phase, or bar-theta value is derived.
2. No numerical gauge coupling, Yukawa eigenvalue, fermion mass, Higgs scale, or electric-dipole moment is predicted.
3. The observed generation count is fixed-boundary target data and is not selected by SYM-4.
4. The project does not solve the strong-CP problem or derive a Peccei-Quinn, axion, Nelson-Barr, or other BSM mechanism.
5. The one-Higgs-doublet theorem does not classify spontaneous CP violation in multi-Higgs, singlet, or composite scalar targets.
6. The project does not supply a complete SMEFT CP basis without a declared operator dimension and matching scale.
7. The project does not derive baryogenesis or the cosmological matter-antimatter asymmetry.
8. A nonzero naive-T-odd correlation is not automatically fundamental T violation.
9. A displayed complex coefficient or mixing-matrix phase is not automatically a physical CP invariant.
10. CPT exactness does not imply exactness of C, P, T, or CP individually.
11. Global-form, line, monopole, higher-group, and non-invertible completion remains assigned to SYM-9.
12. Whole-arc candidate exhaustion and same-scope atlas closure remains assigned to SYM-11.
13. Imported continuum-QFT theorems retain imported status; the executable proof verifies their target specialization and finite normal-form consequences.
14. Skin means representative dependence, not unreality, uselessness, or empirical dispensability.

B.1 Interpretation rule

“Derived”, “witness”, “skin”, “effective”, “obstruction” and “externally fixed” are dependency or status classifications. They do not mean unreal, useless, empirically irrelevant or mathematically optional. Conversely, exactness or physical importance does not establish primitive carrier standing.

Proposition-Level Source Interfaces

The table lists the corpus propositions that directly feed the manuscript-level target and role construction. The Technical Audit Companion contains the complete local source registry and hashes.

Table C.1: Proposition-level source ledger.

ID	Exact statement	Hypotheses	Source	Locator	Target specialization	Limitations	Status
P-AMETRIC-BOUNDARY	Within a fixed standing regime no selector, grading, metric, probability, typicality rule, canonical representative, repair, generator, carrier, or hidden boundary authority can independently bind standing.	Fixed typed domain, standing-instantiated application regime, irreversibility and bivalent admissibility.	The AMetric Boundary as a Unique Regime Invariant Constraint on Admissibility Bearing Construction	Theorems 6.12-6.27, especially Theorem 6.27	SYM-4 target IDs listed by theorem map	Applies only within the frozen same-scope comparison regime; it does not select the local SYM-4 occupant.	imported
P-AOT2-NEUTRINO	Branch orientation and fitted values cannot be imported upstream of invariant neutrino projection.	Orientation-neutral neutrino target.	AOT2 Neutrino Branch Orientation and Prediction Lock Transport v4	Secs. 2-4	SYM-4 target IDs listed by theorem map	Does not derive measured PMNS coordinates.	imported
P-APEX-CLOSURE	Under the complete standing-instantiated application package, the apex standalone closure theorem excludes a second same-domain standing classifier or governance-non-equivalent admissible regime.	All hypotheses of Theorem 6 plus the theorem-specific reference, scope and meta-invariance interfaces.	Unus Solus Possibilis Est	Section 8, Theorem 7 (Apex Standalone Closure Theorem)	SYM-4 target IDs listed by theorem map	Imported terminal consequence only after local singleton closure and exact premise transport.	imported
P-APEX-UNIQUE-INTERIOR	In the standing-instantiated application regime, exactly one admissible interior exists.	Application-regime anchor, intelligibility, irreversibility, admissible composition, scope discipline, meta-invariance, AMetric boundary, bivalence, target fixation and a nonempty standing-positive side.	Unus Solus Possibilis Est	Section 7, Theorem 6 (Uniqueness of the admissible interior)	SYM-4 target IDs listed by theorem map	Imported terminal consequence; it does not generate the local SYM-4 assignment or repair a missing target coordinate.	imported
P-CNF7	Weak-basis, rephasing and exact-degeneracy quotients separate physical phase residues from display coordinates.	Fixed independent mass carriers and mixing target.	CNF7: Mixing Angle and Phase Forcing over Independently	mixing, rephasing and degeneracy sections	SYM-4 target IDs listed by theorem map	No numerical mixing value is derived.	imported
P-DATTA-LONDON-TP	Triple-product correlations separate true weak-phase CP/T information from fake absorptive contributions in B to VV amplitudes.	Fixed transversity amplitudes and CP-conjugate channel.	Datta London	Secs. 2-3	SYM-4 target IDs listed by theorem map	A kinematic T-odd sign alone is not a direct T test.	imported
P-ELITZUR	A local gauge-variant order parameter cannot establish spontaneous breaking of local gauge redundancy.	Local gauge theory.	Elitzur	Phys. Rev. D 12, 3978-3982	SYM-4 target IDs listed by theorem map	Bibliographic primary source; local byte not supplied.	imported
P-FIXED-DOMAIN	Changes of field content, background, boundary, global form, operator category, phase or scale are target updates unless parameterized.	Fixed-domain comparison.	A Fixed Domain Exhaustion Theorem	scope-lock and target-update theorem	SYM-4 target IDs listed by theorem map	Does not forbid enlarged-domain physics.	imported
P-GENCOUNT	Generation count is fixed-boundary envelope data; changing it changes flavor, Yukawa and mixing ledgers.	Fixed Standard Model representation envelope.	Generation Count and Residual Parameters in the Standard Model Representation Regime	Theorem 9.8 and Theorem 18.1	SYM-4 target IDs listed by theorem map	Does not derive the numerical value three.	imported
P-GRIMUS-REBELO-CP	Generalized CP and parity transformations in gauge theories are controlled by automorphisms and compatible field/Yukawa maps.	Declared gauge group, representations and Yukawa sector.	Grimus Rebelo	Secs. 2-5	SYM-4 target IDs listed by theorem map	Target-specific implementation remains required.	imported
P-HIGGS-ENDPOINT	The Higgs bridge carrier is distinct from field, vev, gauge representative, scale, vacuum route and extensions.	One-doublet Higgs structural target.	The Higgs Domain Endpoint: Carrier: Identity: Scale: Vacuum: Extension: and Projection	endpoint theorem package	SYM-4 target IDs listed by theorem map	Does not classify arbitrary extended scalar sectors.	imported
P-HIGGS-VACUUM	Vacuum and metastability are route statuses, and gauge representatives do not independently ground physical CP breaking.	Fixed Higgs carrier, gauge quotient and selected vacuum orbit.	Higgs Vacuum Route Status: Metastability: Lower Branches: and Standing Closure	vacuum route theorem	SYM-4 target IDs listed by theorem map	Multi-Higgs spontaneous CP is a target update.	imported
P-JARLSKOG-1985	The commutator of quark mass matrices supplies a weak-basis invariant CP criterion in the generic nondegenerate three-generation target.	Three generations and nondegenerate quark masses.	Jarlskog	Phys. Rev. Lett. 55, 1039-1042	SYM-4 target IDs listed by theorem map	Degenerate and lower-generation targets require separate treatment.	imported
P-KERNEL-NECESSITY	If a meaningful regime admits non-degenerate construction, the full Admissibility-Standing-Reference-Irreversibility kernel is already in force; no meaningful regime supports non-degenerate construction while lacking one of those functions or a governance-equivalent replacement.	A meaningful regime with non-degenerate determinate construction in the sense of the source definition.	Non Degenerate Construction and the Kernel of Admissibility	p. 20, Theorem 2.34	All closed SYM-4 physical targets after local pre-role determinacy is established.	This imported theorem establishes necessity of the kernel; target-specific determinacy is supplied locally and downstream software cannot revoke the theorem.	imported_theorem
P-NEUTRAL-FCNC	With fixed Standard Model fermion content and gauge representations and one Higgs doublet, unitary mass-basis transformations leave tree-level neutral gauge currents flavor diagonal; tree-level flavor-changing neutral currents are absent.	Standard Model field content, fixed gauge representations, a single Higgs doublet, unitary Yukawa diagonalization, and the tree-level renormalizable neutral-current operator locus.	L1: Paper 10 Absence of Tree Level Flavor Changing Neutral Currents	p. 2, Theorem 1	F-WEAK-NEUTRAL-CURRENT-TREE-3GEN and the integrated SYM-4 ledger assembly.	Tree-level renormalizable Standard Model scope only; loop-induced and higher-dimension flavor-changing neutral currents are distinct targets.	imported_theorem_with_local_specialization
P-QCD-THETA-C-EVEN	The QCD theta density and quark mass term are C-even in the declared strong-interaction target, while generic nonzero bar-theta can violate P, T and CP.	Fixed strong-sector field content, charge-conjugation convention, mass matrix and theta target.	Anomalies as Structural Obstructions	Chs. 20 and 22; strong-CP handoff	SYM-4 target IDs listed by theorem map	Special theta values and global form remain target-indexed.	imported

continued

ID	Exact statement	Hypotheses	Source	Locator	Target specialization	Limitations	Status
P-STRONGCP-STRUCTURA	The strong phase is the anomaly-invariant combination $\bar{\text{bar}}\text{-}\theta = \theta + \arg \det M$ in the declared convention.	Fixed chirality, phase and topology conventions.	Structura Ex Necessitate Standardis Modelis	strong-CP invariant lemma and ledger	SYM-4 target IDs listed by theorem map	Special theta points require vacuum/global-form data.	imported
P-SYM0-STATUS	Discrete-symmetry claims require independent target, quotient, quantum, vacuum, boundary, scale and role coordinates.	Fixed target and complete certificate.	The Symmetry Status(1)	Chs. 3-5 and benchmark dossiers	SYM-4 target IDs listed by theorem map	Calibration universe, not whole-arc exhaustion.	imported
P-SYM2-CPT	CPT is a theorem consequence and antiunitary physical action under the declared Jost/Wightman premise packet.	Local relativistic QFT, positive metric, spectrum, locality and the registered formulation.	Spacetime Symmetry as Domain Anchor	Chs. 7, 13 and CPT appendices	SYM-4 target IDs listed by theorem map	Other CPT formulations require explicit bridges.	imported
P-SYM3-ANOMALY	Gauge inconsistency, anomalous global response, theta transport and matching are distinct target-indexed anomaly records.	Reference-complete anomaly target.	Anomalies as Structural Obstructions	Chs. 6-12, 20 and 22	SYM-4 target IDs listed by theorem map	Global-form and defect completion remains SYM-9.	imported
P-ZHOU-MAJORANA	Majorana CP conservation requires target-relative weak-basis invariant conditions beyond ordinary oscillation quartets.	Three-generation Majorana mass target.	Zhou Weak Basis	main invariant theorem and examples	SYM-4 target IDs listed by theorem map	Exact condition count depends on target assumptions.	imported
P-LOCAL-DETERMINACY	Each closed target has one target-fixed standing-positive completed act, one target-fixed standing-negative completed act, a nonempty and nonuniversal gate, an irreversible disposition, and a reportable local endpoint before role occupation.	frozen 4.4.1-rc1 primitive and proof artifacts	SYM-4 4.4.1-rc1 executable certificate	audit/apex_application_index.json	all declared 4.4.1-rc1 targets as indexed	finite theorem; imported QFT truth remains external	locally_proved
P-LOCAL-DENIAL	Executed interface denials generate target-relative query-loss profiles and governance classes.	frozen 4.4.1-rc1 primitive and proof artifacts	SYM-4 4.4.1-rc1 executable certificate	audit/denial_transitions.json	all declared 4.4.1-rc1 targets as indexed	finite theorem; imported QFT truth remains external	locally_proved
P-LOCAL-INDEPENDENCE	Burden and candidate primitives are process- and schema-isolated until operational eligibility.	frozen 4.4.1-rc1 primitive and proof artifacts	SYM-4 4.4.1-rc1 executable certificate	audit/burden_independence.json	all declared 4.4.1-rc1 targets as indexed	finite theorem; imported QFT truth remains external	locally_proved
P-LOCAL-CONSTRUCTOR	The complete formula-bearing constructor registry reaches a finite fixed point with derivation paths.	frozen 4.4.1-rc1 primitive and proof artifacts	SYM-4 4.4.1-rc1 executable certificate	audit/constructor_fixed_point.json	all declared 4.4.1-rc1 targets as indexed	finite theorem; imported QFT truth remains external	locally_proved
P-LOCAL-PRESENTATION	Executable forward/inverse/composition maps generate the witnessed presentation groupoid.	frozen 4.4.1-rc1 primitive and proof artifacts	SYM-4 4.4.1-rc1 executable certificate	audit/presentation_groupoid_axioms.json	all declared 4.4.1-rc1 targets as indexed	finite theorem; imported QFT truth remains external	locally_proved
P-LOCAL-ELIGIBILITY	Every same-fiber role-residue pair receives an operational query-restoration disposition.	frozen 4.4.1-rc1 primitive and proof artifacts	SYM-4 4.4.1-rc1 executable certificate	data/eligibility_matrix.csv	all declared 4.4.1-rc1 targets as indexed	finite theorem; imported QFT truth remains external	locally_proved
P-LOCAL-COMPATIBILITY	Every complete eligible map is reassembled and checked against target queries and cross-object equations.	frozen 4.4.1-rc1 primitive and proof artifacts	SYM-4 4.4.1-rc1 executable certificate	audit/compatibility_summary.json	all declared 4.4.1-rc1 targets as indexed	finite theorem; imported QFT truth remains external	locally_proved
P-LOCAL-AUTOMORPHISM	Full colored-certificate automorphisms determine assignment orbits.	frozen 4.4.1-rc1 primitive and proof artifacts	SYM-4 4.4.1-rc1 executable certificate	data/assignment_orbits.csv	all declared 4.4.1-rc1 targets as indexed	finite theorem; imported QFT truth remains external	locally_proved
P-LOCAL-NORMALFORM	Every closed fiber and the separately typed assembly have one assignment orbit.	frozen 4.4.1-rc1 primitive and proof artifacts	SYM-4 4.4.1-rc1 executable certificate	audit/proof_results.json	all declared 4.4.1-rc1 targets as indexed	finite theorem; imported QFT truth remains external	locally_proved
P-LOCAL-REFINEMENT	Mutation, ablation, holdout and rival tests reopen load-bearing failures.	frozen 4.4.1-rc1 primitive and proof artifacts	SYM-4 4.4.1-rc1 executable certificate	audit/target_coordinate_mutations.json	all declared 4.4.1-rc1 targets as indexed	finite theorem; imported QFT truth remains external	locally_proved
P-LOCAL-CONTINUATION	Frontiers and successor targets are explicit and excluded from the closed quantifier.	frozen 4.4.1-rc1 primitive and proof artifacts	SYM-4 4.4.1-rc1 executable certificate	data/target_registry.csv	all declared 4.4.1-rc1 targets as indexed	finite theorem; imported QFT truth remains external	locally_proved
P-LOCAL-EVIDENCE	Every theorem has a proposition-level target-matched evidence path.	frozen 4.4.1-rc1 primitive and proof artifacts	SYM-4 4.4.1-rc1 executable certificate	data/theorem_evidence.csv	all declared 4.4.1-rc1 targets as indexed	finite theorem; imported QFT truth remains external	locally_proved
P-LOCAL-RECONSTRUCTION	A code-separated implementation independently replays the canonical target-level role/residue counts, incidence counts, complete assignment counts, compatibility dispositions, lawful automorphism counts, and orbit counts from the frozen point-target packets. It does not independently reconstruct the primary raw-constructor fixed point or witnessed raw-to-canonical presentation quotient.	frozen 4.4.1-rc1 primitive and proof artifacts	SYM-4 4.4.1-rc1 executable certificate	audit/independent_reconstruction.json	all declared 4.4.1-rc1 targets as indexed	finite theorem; imported QFT truth remains external	locally_proved
P-LOCAL-DOWNSTREAM	Downstream exports preserve target, theorem and successor discipline.	frozen 4.4.1-rc1 primitive and proof artifacts	SYM-4 4.4.1-rc1 executable certificate	data/downstream_interfaces.csv	all declared 4.4.1-rc1 targets as indexed	finite theorem; imported QFT truth remains external	locally_proved
P-LOCAL-AUDIT	The executable audit suite passes under the frozen gate architecture.	frozen 4.4.1-rc1 primitive and proof artifacts	SYM-4 4.4.1-rc1 executable certificate	audit/audit_tests.json	all declared 4.4.1-rc1 targets as indexed	finite theorem; imported QFT truth remains external	locally_proved

C.1 Authority rule

A source claim is admitted only when its target hypotheses match the receiving record and its version is fixed. Citation identity, local file integrity and theorem truth remain separate coordinates. A local SHA-256 match establishes source-byte identity, not the truth of the imported proposition.

Notation and Controlled Vocabulary

Symbol or value	Meaning
$D_4^\star$	Indexed family containing the integrated target, closed projection targets, frontiers and successor transports.
D_λ	One Reference-complete target fiber.
K_{D_λ}	Admissibility–Standing–Reference–Irreversibility kernel instantiated in the target.
B_λ	Raw target-denial burdens.
$R_\lambda = B_\lambda / \sim_{\text{gov}}$	Canonical target-role architecture after governance-equivalent burdens are identified.
O_λ	Role-blind parent physical-object registry.
$\mathcal{H}_4$	Registered physical constructor grammar.
F_λ	Fixed-point physical facet universe.
$X_\lambda = F_\lambda / \sim_{\text{pres}}$	Presentation-quotiented physical residues.
E_λ	Complete role-residue eligibility relation.
J_λ	Relational compatibility predicate on complete eligible maps.
$\Gamma_\lambda^{\text{lawful}}$	Full-invariant-preserving automorphism group.
$\mathcal{N}_4(D_\lambda)$	Groupoid of compatible complete assignments and lawful automorphisms.
$\bar{\theta}$	Anomaly-invariant strong-CP phase coordinate in the convention $\bar{\theta} = \theta + \arg \det M_q$.
Generalized CP	Antiunitary CP operation possibly composed with an allowed unitary map in flavor or degenerate subspaces.
Presentation witness	Invertible map preserving the complete target, physical quotient, invariant, continuation and failure tuple.
Physical action	Transformation that changes physical states, operators, charges, sectors or selection rules after quotient.
Skin	Variation that changes only presentation after all physical and standing-bearing content is fixed.
Derived witness	Current, invariant, theorem, anomaly or diagnostic consequence that records structure without independently grounding it.
Bridge/fixation	Structure mediating or fixing a phase, response or sector connection, including the Higgs bridge carrier.
Target update	Change of field content, background, global form, topology, vacuum, operator category, neutrino convention or scale scope.
Open burden	Precisely typed missing information preventing the target from entering the closed quantifier.

Bibliography

- Amos Jay Maley. Non-degenerate construction and the kernel of admissibility: An exhaustion theorem for standing, reference, and irreversibility. AASC corpus manuscript, 2026a.
- Amos Jay Maley. A fixed-domain exhaustion theorem: Standing, admissibility, and the determination of consequence under typed boundary discipline. AASC corpus manuscript, 2026b.
- Amos Jay Maley. The symmetry status calculus: Constraint, redescription, physical action, and standing. SYM-0 publication release, 12 July 2026, 2026c.
- Amos Jay Maley. Symmetry as constraint, redescription, and physical action: A detailed research arc for classifying the symmetries of the standard model within the carrier/skin program. AASC corpus manuscript, 11 July 2026, 2026d.
- Amos Jay Maley. Unus solus possibilis est: Corpus closure and uniqueness of the admissible interior. AASC corpus manuscript, 2026e.
- Amos Jay Maley. Role-occupancy closure under admissibility: Why role-cardinality matching is only the cardinality shadow of compatible realization. AASC corpus manuscript, 2026f.
- Amos Jay Maley. Generation count and residual parameters in the standard model representation regime: A residual-exhaustion theorem for fixed-boundary data. AASC corpus manuscript, May 2026, 2026g.
- Amos Jay Maley. The higgs bridge carrier: Kernel consequences for electroweak mass response. Paper 1 of the Higgs Structural Endpoint Arc, 2026h.
- Amos Jay Maley. Endpoint on higgs structural identity: Bridge-carrier occupation and no fifth identity ground. Paper 2 of the Higgs Structural Endpoint Arc, 2026i.
- Amos Jay Maley. No primitive higgs scale: Carrier-indexed scale standing and hierarchy orientation. Paper 3 of the Higgs Structural Endpoint Arc, 2026j.
- Amos Jay Maley. Higgs vacuum route status: Metastability, lower branches, and standing closure. Paper 4 of the Higgs Structural Endpoint Arc, 2026k.
- Amos Jay Maley. The higgs extension classifier: Extra scalars and same-scope carrier preservation. Paper 5 of the Higgs Structural Endpoint Arc, 2026l.
- Amos Jay Maley. The higgs domain endpoint: Carrier, identity, scale, vacuum, extension, and projection. Paper 6 of the Higgs Structural Endpoint Arc, 2026m.
- Stephen L. Adler. Axial-vector vertex in spinor electrodynamics. *Physical Review*, 177:2426–2438, 1969. doi: 10.1103/PhysRev.177.2426.
- J. S. Bell and R. Jackiw. A pcac puzzle: $\pi^0 \rightarrow \gamma\gamma$ in the sigma model. *Il Nuovo Cimento A*, 60:47–61, 1969.
- Kazuo Fujikawa. Path-integral measure for gauge-invariant fermion theories. *Physical Review Letters*, 42:1195–1198, 1979. doi: 10.1103/PhysRevLett.42.1195.
- Amos Jay Maley. Structura ex necessitate standardis modelis: A relative closure theorem for the standard-model bookkeeping interior under fixed load, fixed domain, and certificate-maximal same-domain transport. AASC corpus manuscript, 2026n.
- Amos Jay Maley. Anomalies as structural obstructions: Gauge inconsistency, global response, inflow, matching, and normal-form closure in the standard model. SYM-3 combined archival reference, July 2026, 2026o.

- Ziro Maki, Masami Nakagawa, and Shoichi Sakata. Remarks on the unified model of elementary particles. *Progress of Theoretical Physics*, 28:870–880, 1962. doi: 10.1143/PTP.28.870.
- J. Schechter and J. W. F. Valle. Neutrino masses in $su(2) \times u(1)$ theories. *Physical Review D*, 22:2227–2235, 1980. doi: 10.1103/PhysRevD.22.2227.
- S. M. Bilenky, J. Hosek, and S. T. Petcov. On oscillations of neutrinos with dirac and majorana masses. *Physics Letters B*, 94:495–498, 1980.
- Amos Jay Maley. Neutrino branch orientation and prediction-lock transport. AOT2 v4, AASC corpus manuscript, 2026p.
- Amos Jay Maley. Neutrino oscillation as the unique admissible reconstruction of flavor under noncommuting propagation. AASC corpus manuscript, 2026q.
- Nicola Cabibbo. Unitary symmetry and leptonic decays. *Physical Review Letters*, 10:531–533, 1963. doi: 10.1103/PhysRevLett.10.531.
- Makoto Kobayashi and Toshihide Maskawa. Cp-violation in the renormalizable theory of weak interaction. *Progress of Theoretical Physics*, 49:652–657, 1973. doi: 10.1143/PTP.49.652.
- Cecilia Jarlskog. Commutator of the quark mass matrices in the standard electroweak model and a measure of maximal cp nonconservation. *Physical Review Letters*, 55:1039–1042, 1985. doi: 10.1103/PhysRevLett.55.1039.
- T. D. Lee and C. N. Yang. Question of parity conservation in weak interactions. *Physical Review*, 104:254–258, 1956. doi: 10.1103/PhysRev.104.254.
- C. S. Wu, E. Ambler, R. W. Hayward, D. D. Hoppes, and R. P. Hudson. Experimental test of parity conservation in beta decay. *Physical Review*, 105:1413–1415, 1957. doi: 10.1103/PhysRev.105.1413.
- J. H. Christenson, J. W. Cronin, V. L. Fitch, and R. Turlay. Evidence for the 2π decay of the k_2^0 meson. *Physical Review Letters*, 13:138–140, 1964. doi: 10.1103/PhysRevLett.13.138.
- CPLEAR Collaboration. First direct observation of time-reversal non-invariance in the neutral-kaon system. *Physics Letters B*, 444:43–51, 1998. doi: 10.1016/S0370-2693(98)01356-2.
- BaBar Collaboration. Observation of time-reversal violation in the b^0 meson system. *Physical Review Letters*, 109: 211801, 2012. doi: 10.1103/PhysRevLett.109.211801.
- Shmuel Elitzur. Impossibility of spontaneously breaking local symmetries. *Physical Review D*, 12:3978–3982, 1975. doi: 10.1103/PhysRevD.12.3978.
- Amos Jay Maley. Spacetime symmetry as domain anchor: Poincare, lorentz, translation, and cpt. SYM-2 research article, 15 July 2026, 2026r.
- Hilary Greaves and Teruji Thomas. On the cpt theorem. *Studies in History and Philosophy of Modern Physics*, 45: 46–65, 2014. doi: 10.1016/j.shpsb.2013.10.001.
- Amos Jay Maley. The standalone standard model structure from first principles: An aasc derivation of gauge-product, representation, higgs/bridge, yukawa, charge, anomaly, vacuum, and parameter-status structure. AASC corpus manuscript, 2026s.
- Amos Jay Maley. Mixing-angle and phase standing over independently standing mass carriers. MGN6, AASC corpus manuscript, 2026t.
- Amos Jay Maley. Cnf7: Mixing-angle and phase forcing over independently standing mass carriers. AASC corpus manuscript, 2026u.

- Amos Jay Maley. Cp-like phases as admissible holonomy classes. SM-PARAM-F, AASC corpus manuscript, 2026v.
- Amos Jay Maley. Discrete holonomy exactness: Phase residues, loop classes, finite transport invariants, and nonlocal exact outputs. EX7, AASC corpus manuscript, 2026w.
- Amos Jay Maley. Yukawa compatibility and parameter-survival classification over the canonical bridge carrier. MGN4, AASC corpus manuscript, 2026x.
- Amos Jay Maley. Gauge architecture and gauge skin in the standard model. SYM-1 publication release v9.0.0, 13 July 2026, 2026y.
- David Tong. Line operators in the standard model. *Journal of High Energy Physics*, 07:104, 2017. doi: 10.1007/JHEP07(2017)104.
- Joe Davighi, Ben Gripaios, and Nakarin Lohitsiri. Global anomalies in the standard model(s) and beyond. *Journal of High Energy Physics*, 07:232, 2020. doi: 10.1007/JHEP07(2020)232.
- Alakabha Datta and David London. Triple-product correlations in $b \rightarrow v_1 v_2$ decays and new physics. *International Journal of Modern Physics A*, 19:2505–2544, 2004.
- T. D. Lee. A theory of spontaneous t violation. *Physical Review D*, 8:1226–1239, 1973. doi: 10.1103/PhysRevD.8.1226.
- Amos Jay Maley. Boundary-reconstruction and mixing consequences of the standing neutrino ratio interval. NNR6, AASC corpus manuscript, 2026z.
- Walter Grimus and M. N. Rebelo. Automorphisms in gauge theories and the definition of cp and p. *Physics Reports*, 281:239–308, 1997.